Impacts of Secondary School in Kenya after Two Decades

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Abstract

This paper studies the long-run effects of access to secondary school in Kenya – where admission depended on a cutoff score on a national exam – using a regression discontinuity design and a panel dataset that follows individuals for nearly two decades after the exam. Scoring above the cutoff raises secondary school completion by 18 percentage points, with similar increases for women and men. Greater access to secondary schooling leads to substantial improvements in cognition, with gains of 0.2–0.3 standard deviations into midlife. Realized fertility also declines substantially and persistently among men (by about 0.8 children on average) but, perhaps surprisingly, not for women. There is suggestive evidence that benefits may extend to the next generation, as the children of female respondents who scored above the exam cutoff perform better on tests of academic skills.

JEL codes: I21, J31, O12, O15.

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1 Introduction

The focus of global education policy has shifted toward secondary schooling in recent years as part of the Sustainable Development Goals. About half of the countries in Sub-Saharan Africa now offer fee-free public secondary school (Garlick, 2019; Ferreira and Sandholtz, 2026); for instance, in Kenya, the setting of this study, the government abolished tuition fees in public secondary schools in 2008 (Ohba, 2011; Brudevold-Newman, 2021). Yet credible evidence on the long-run effects of secondary schooling in low- and middle-income countries (LMICs) remains scarce, because secondary school expansions typically reach entire cohorts at once and few studies follow students far into adulthood to measure effects on life outcomes.

This paper studies the long-run effects of access to secondary school in Kenya. Under Kenya's former school system, admission to government secondary schools depended on scores on the Kenya Certificate of Primary Education (KCPE).[1] The probability of admission rose sharply at a particular score, which makes it possible to compare students who scored just above versus just below the cutoff. Ozier (2018) used this discontinuity to study the same cohort in early adulthood and found gains in vocabulary and reasoning, a reduction in low-skill self-employment among men, and a reduction in teen pregnancy for women. This paper follows the same individuals for about 15 more years, into their late 30s and early 40s.

The analysis links administrative KCPE exam records to the Kenya Life Panel Survey (KLPS), which has followed approximately 7,500 individuals from rural western Kenya since they were enrolled in primary school in 1998 through 2026.[2] Additional rounds of data collection, combined with improved matching to the exam records, expands the analysis sample by about 1,000 individuals relative to Ozier (2018), to 3,162 respondents. The most recent round of data collection usefully administered an adaptation of the Harmonized Cog-

[1] Kenya has since moved from the so-called "8–4–4" system to a competency-based curriculum (Akala, 2021); the last KCPE was administered in 2023 (Kalekye, 2023). Placement into senior secondary school is now based on the Kenya Junior School Education Assessment, which pupils first took in 2025 at the end of junior secondary school (Ayega, 2025).

[2] Accounting for the two-stage tracking approach of the KLPS, effective survey rates have generally been 80 to 85% in each round (Hsu et al., 2026).

nitive Assessment Protocol (HCAP), a cognitive battery of assessments covering memory, language, executive function, orientation, and visuospatial ability (Gross et al., 2025). The survey rounds also record fertility, marriage, earnings, and household consumption. The most recent round also assessed the language, math, executive function, and reasoning skills of up to two children aged 8 to 14 per respondent, allowing for an examination of effects on the next generation.

Crossing the KCPE cutoff substantially increases educational attainment, raising the probability of completing secondary school by 18 percentage points (from a below-cutoff mean of 30 percent) and adding about one year of schooling on average. This is similar to the 15 percentage point short-run increase that Ozier (2018) estimated in the original sample. The increase in secondary school completion is similar for women (22 percentage points) and men (21 percentage points). Many regression discontinuity (RD) studies of school admission compare students admitted to more versus less selective schools, nearly all of whom attend school, and therefore estimate the effect of school quality (Jackson, 2010; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Lucas and Mbiti, 2014). In this study's setting, by contrast, about a third of students who scored just below the cutoff never attend any secondary school (and many only attend some before dropping out), so the comparison captures, in large part, the effect of access to additional secondary schooling itself rather than only the effect of school quality.[3]

In a new finding, the data show that crossing the exam score cutoff also leads to large and lasting gains in individual cognitive performance. At the time of the cognitive assessment, when respondents were 38 years old on average (10-90 age range 34 to 41), those just above the cutoff score 0.24 standard deviation units higher than those just below it on a general cognition factor that summarizes performance across all five main cognitive domains. Among the separate domains, the gain is largest for memory (0.26 standard deviations, significant at the 95% level), the estimated effect on visuospatial ability is 0.21 standard deviations (sig-

[3]Specifically, among students within 20 exam score points below the cutoff, 31% never attended any secondary school (36% of women and 26% of men).

nificant at 90%), while the estimates for language, executive function, and orientation are not statistically significant. These effects are large given that the cutoff changes secondary completion status for only roughly one in six students. Scaled by the effect on years of schooling, they imply that each additional year of schooling raises general cognitive performance by about 0.25 standard deviations. Together with the gains in vocabulary and reasoning that Ozier (2018) found in respondents' early 20s, these results show that the effects of secondary schooling on cognitive performance emerge in the short-run and remain sizable nearly two decades after the exam. Despite widespread concern that expanding access to school in LMICs often produces little actual learning (Pritchett, 2013; World Bank, 2018), this finding indicates that secondary schooling in this setting generated lasting individual human capital.

Despite these substantial gains in cognition, there is limited evidence of increases in earnings or household consumption, although measurement issues may be playing a role. In data pooled over four survey rounds from respondents' early 20s to their late 30s (on average), the estimated effect on annual non-farm earnings is near zero (−87 dollars relative to a below-cutoff mean of 2,197 dollars, 2024 USD PPP), and the effect on per capita household consumption is also small and not statistically significant. These estimates are relatively imprecisely estimated, however, reflecting well-known difficulties measuring informal market outcomes in LMIC settings. It is worth noting that the 95% confidence interval for earnings gains, which runs from 36% below to 22% above the mean, includes substantial gains. The scarcity of formal wage jobs in this setting, particularly for women, may limit the earnings returns to additional schooling, since returns to education in Africa are often thought to be concentrated in these jobs (Rankin et al., 2010; Byambaa et al., 2025).

Most research on schooling and family formation focuses on women, but in this setting the long-run decline in fertility occurs among men. By their late 30s, men just above the exam cutoff have 0.8 fewer children than men just below it, a 24% decline. The gap emerges in men's mid-20s and widens over time. Women's fertility, in contrast, is unchanged by their

late 30s – even though the increase in schooling at the cutoff is similar for women and men – and the estimates are relatively precise, ruling out declines larger than 0.56 children (at 95% confidence). Together with the reduction in teen pregnancy found by Ozier (2018), this pattern is consistent with delayed childbearing rather than fewer births. Desired family size as stated in surveys also falls by about 0.4 children among women in each survey round, while the estimates for men are not statistically significant. Note that the probability of marriage and the number of marriages do not change at the cutoff for either gender, and the observed demographic and socioeconomic characteristics of spouses are also similar above and below the cutoff, suggesting that more schooling did not lead to sharp changes in marriage patterns.

There is also suggestive evidence that the gains in human capital pass to the next generation, though only through mothers. Children aged 8 to 14 whose mothers scored just above the cutoff perform 0.36 standard deviation units better on an index of academic skills, with the largest gain in language skills (0.41 standard deviations). The children of male respondents show no gains in academic skills, and the difference between the female versus male effects estimates is significant at the 90% level. Because the increase in schooling at the cutoff is similar for women and men, this difference does not simply reflect larger schooling gains for women. These estimated effects on academic skills is significant at the 90% level, however, and the set of children observed (within the tested birth cohorts) depends partly on fertility timing, which itself could respond to the cutoff, so these results should be viewed as suggestive rather than definitive. The body of causal evidence on whether parents' schooling affects their children's skills is quite limited, particularly in LMICs, where most estimates come from expansions of primary schooling (rather than secondary schooling, as in this study). Although suggestive, these findings on the next generation indicate that evaluations focused only on students' own life outcomes may understate the long-run returns to girls' secondary schooling operating through other channels (Duflo et al., 2026).

This paper contributes to three literatures. First, it adds to the limited causal evidence on the long-run effects of secondary schooling in LMICs. The closest study is a randomized

evaluation of secondary school scholarships in Ghana, which followed students into their early 30s (Duflo et al., 2025, 2026). The Ghana scholarships went to students who had been admitted to secondary school but could not afford to enroll, so that study measures the effect of removing a financial barrier to attendance. In Kenya, admission itself depended on the KCPE score, so the estimates here capture effects for students of average academic performance whose access to secondary school depended on a few points on the exam. Despite these differences in the populations affected, the findings are similar in several respects. Both studies find gains in measured skills, little evidence of earnings gains among men, and no significant effect on the number of children women have by the time of the latest follow-up. Both also find some evidence of gains for the children of women but not for the children of men. The current study follows respondents further into adulthood, which makes it possible to measure cognition into the late 30s and early 40s, as well as the academic skills of older children, aged 8 to 14 (in contrast to the Ghana study, which measures cognitive development only through age 7). The longer follow-up also reveals a substantial decline in men's fertility, which is difficult to assess in Ghana, where men in the control group had fewer than one child on average by about age 30, likely far below their lifetime number of children. A follow-up of a cash transfer experiment in Malawi similarly finds that conditional transfers to girls who had dropped out of school had lasting effects on schooling, marriage, and fertility but not on earnings (Baird et al., 2011, 2019).

The long follow-up period in the current study also matters for understanding how schooling affects fertility. Most studies of education and family formation in LMICs examine marriage and childbearing during adolescence or early adulthood (Breierova and Duflo, 2004; Osili and Long, 2008; Chicoine, 2012; Duflo et al., 2015; Kırdar et al., 2018; Keats, 2018; Brudevold-Newman, 2021; Kazibwe and Li, 2025). Effects measured at these ages may reflect changes in the timing of births rather than in the number of children ultimately born. In Turkey, for example, the effects of a compulsory schooling reform on teenage marriage and births disappear by the late teens (Kırdar et al., 2018). Women in this Kenyan cohort show

a similar pattern. Although Ozier (2018) documented a reduction in teenage pregnancy, women who scored above the cutoff show no detectable decline in the number of children they have by their late 30s, which is closer to completed lifetime fertility. This finding is consistent with evidence from compulsory schooling reforms in Europe (Monstad et al., 2008; Fort et al., 2016), although a stipend program in Bangladesh did reduce fertility (Hahn et al., 2018). Among men, by contrast, the decline in fertility emerges in their mid-20s and widens throughout their 30s. A study ending in early adulthood would therefore emphasize delayed childbearing among women while missing much of the decline in men's fertility.

Second, this paper shows that the cognitive gains from secondary schooling in Kenya persist into midlife. Most causal evidence on schooling and later-life cognition comes from compulsory schooling reforms in Europe and the United States, which find lasting improvements in memory (Glymour et al., 2008; Banks and Mazzonna, 2012; Schneeweis et al., 2014). Notably, the largest gains in the current Kenyan setting are also in the memory domain. A meta-analysis estimates that each additional year of schooling raises IQ by 1 to 5 points, or approximately 0.07 to 0.33 standard deviations (Ritchie and Tucker-Drob, 2018). The estimate here of about 0.25 standard deviations per year of schooling falls squarely within this range. Evidence from LMICs on this issue remains limited and mixed, with most studies measuring skills shortly after schooling or examining the effects of primary education (Filmer and Schady, 2014; Ozier, 2018; Duflo et al., 2025; Amin et al., 2025). These findings are particularly relevant for Sub-Saharan Africa, where the number of older adults is projected to triple over the next three decades (Duhon et al., 2023) and low educational attainment is considered a leading modifiable risk factor for dementia (Livingston et al., 2020).

Third, the results contribute to the relatively small literature on the causal intergenerational effects of parental education. Studies of schooling reforms, school entry rules, and college access in Europe and the United States find mixed effects on children's outcomes (Oreopoulos et al., 2006; Currie and Moretti, 2003; McCrary and Royer, 2011; Carneiro et al., 2013), with some finding gains only from mothers' schooling (Black et al., 2005; Lundborg

et al., 2014). In LMICs, causal evidence comes largely from expansions of primary schooling and other education reforms (Breierova and Duflo, 2004; Akresh et al., 2023; Güneş, 2015; Grépin and Bharadwaj, 2015; Agüero and Ramachandran, 2020; Wantchekon et al., 2015; Andrabi et al., 2012) and from childhood health interventions in the KLPS cohort (Walker et al., 2023; Duhon et al., 2024). The closest evidence on secondary schooling in Sub-Saharan Africa comes from the Ghana experiment discussed above, which measured children's cognitive development up to age 7 (Duflo et al., 2026). The current paper extends that evidence to academic skills among children aged 8 to 14, which is arguably more meaningful since it is likely to directly influence important schooling outcomes.

The remainder of the paper proceeds as follows. Section 2 describes the Kenyan secondary school system, and Section 3 describes the KLPS and the administrative exam records. Section 4 presents the RD design and assesses its validity. Section 5 reports effects on education, cognition, fertility and family formation, earnings and consumption, and respondents' children, and the final section concludes.

2 Background: Secondary Schooling in Kenya

The cohorts in this study were educated under Kenya's 8–4–4 system, which consisted of eight years of primary school (Standards 1 to 8) followed by four years of secondary school (Forms 1 to 4), and then four years of university instruction. At the end of primary school, pupils took the Kenya Certificate of Primary Education (KCPE), a national examination used to determine admission to government secondary schools (Ozier, 2018). A score of 250 out of 500 was widely regarded as a "passing" grade (Akolo, 2008; Ozier, 2018).[4]

Yet admission thresholds varied somewhat across schools and by gender. More selective secondary schools required higher scores (Lucas and Mbiti, 2014; Ozier, 2018). In a 2009 survey of secondary schools in the study area, no school reported an admission cutoff below

[4]For pupils who took the KCPE in 1999–2000, the examination was rescaled and scored out of 700 points. These scores are rescaled back to the 500-point scale used in subsequent years by multiplying by 5/7.

250 for boys, but 7 of 18 schools reported a cutoff below 250 for girls (Ozier, 2018). These thresholds were not strictly binding: students who scored above 250 were not guaranteed admission since the number of qualifying candidates in a given year could exceed the available places (Aduda, 2008; Akolo, 2008; Ozier, 2018). Those who were not admitted could repeat Standard 8 and retake the KCPE, enroll in a private secondary school, or instead pursue vocational training (Ozier, 2018).

Moreover, even students admitted to government secondary schools could not always afford to attend. Before 2008, tuition and other fees at even the least selective government secondary schools amounted to hundreds of dollars per year (Ozier, 2018), a sizable share of annual income for many rural households. In 2008, following the 2007 presidential election, the government abolished tuition fees in public secondary schools, although families continued to face other schooling costs (Ohba, 2011). The cohorts in this study took the KCPE between 1999 and 2006, before the fee reform, when both admission rules and fees limited many children's access to secondary school.

Although there was no single admission cutoff for all students, the probability of entering government secondary school rose sharply at particular scores and this study exploits that variation. Following Ozier (2018), the analysis identifies these cutoffs separately by gender and uses the same threshold values, 234 points for girls and 251 points for boys (Section 4). Students who just cleared these thresholds were far more likely to gain admission to the least selective government schools, while many who just missed them did not attend secondary school at all (as shown below). The comparison at the cutoff therefore differs from many RD studies of school admission: scoring above the cutoff changes whether some students attend secondary school at all, although for students who would have attended anyway it may also change which school they attend. Because students who missed the cutoff could retake the exam, the analysis uses each student's first score, in the spirit of an intention to treat (ITT) analysis.

3 Data

3.1 Kenya Life Panel Survey

The primary data source for this study is the Kenya Life Panel Survey (KLPS), a longitudinal survey that has followed approximately 7,500 individuals who attended 75 primary schools in Busia District, in rural western Kenya, in 1998 (Hsu et al., 2026). These schools took part in the Primary School Deworming Project, a randomized evaluation of school-based deworming (Miguel and Kremer, 2004). The KLPS has completed four survey rounds (2003–2005, 2007–2009, 2011–2014, and 2017–2022), with effective tracking rates of 80 to 85% in each round, and a fifth round (KLPS-5) began in 2023 and is ongoing (Hsu et al., 2026).[5]

Across rounds, the KLPS has collected detailed information on educational attainment, employment, earnings, household consumption, marriage, and fertility. Because it follows respondents from childhood into midlife, the survey makes it possible to trace how the effects of secondary schooling evolve over the life course, including whether early changes in employment and fertility persist as respondents grow older.

KLPS-5 added a detailed assessment of adult cognitive function in the Cognitive Plus (C+) module. The module administered an adaptation of the Harmonized Cognitive Assessment Protocol (HCAP), a battery of neuropsychological tests developed for the U.S. Health and Retirement Study and now used in comparable studies of aging across multiple countries. The KLPS version was adapted to the Kenyan context, and Gross et al. (2025) show that the Kenyan version reliably measures cognition in this population. The assessment covers five domains: orientation, memory, executive function, language, and visuospatial ability.[6] The

[5]KLPS-5 data are collected through several survey modules, each fielded in two representative waves. This paper uses the Cognitive Plus (C+) module, the Integrated (I+) module, and the Kids Assessments and Primary Caregiver (PC) modules, all of which are complete. KLPS-5 earnings come from the C+ module. The Earnings Plus (E+) module, which collects more detailed earnings and consumption data, is still ongoing.

[6]Orientation is assessed with questions on time and place. Memory tests include the CERAD 10-Word List Learning task, the East Boston Memory Test, and a Logical Memory test. Executive function tests include Raven's Progressive Matrices, the Token Test, Go/No-Go, and digit span. Language tests include animal fluency and object naming, and visuospatial tests include CERAD Constructional Praxis, copying interlocking pentagons, and clock drawing (Gross et al., 2025).

analysis here principally uses the general cognition and domain-specific factor scores that Gross et al. (2025) construct using confirmatory factor analysis, standardized to have mean zero and standard deviation one in the full KLPS-5 cognition sample. These assessments provide a more comprehensive measure of cognitive function than the vocabulary and reasoning tests used by Ozier (2018) in early adulthood, making it possible to examine whether cognitive gains are still evident nearly two decades after respondents took the KCPE. In the main analysis sample, 2,451 respondents completed the cognition module.

KLPS-5 also gathered information on respondents' children, which makes it possible to measure outcomes for the next generation. Up to two biological children of each respondent, one in a younger age range and one in an older age range, were selected at random in the previous survey round (Hsu et al., 2026).[7] At the KLPS-5 assessment, the children were aged 8 to 14. The children took tests of vocabulary, verbal fluency, reading, mathematics, working memory, and nonverbal reasoning.[8] Test scores are standardized separately by age group and survey wave, and combined into four indices measuring language, math and spatial ability, executive function, and fluid intelligence. The main outcome is an academic index combining language and math. Because these children are of school age, their performance in reading and mathematics can be measured alongside broader cognitive skills. The Primary Caregiver module gathered in KLPS-5 also records each child's behavior, health, home learning environment, and schooling investments, which are combined into four additional indices. The main analysis sample includes 2,045 assessed children.

3.2 Administrative Data

The KLPS does not record official exam scores, and self-reported KCPE scores are often misreported (Ozier, 2018). Respondents are therefore linked to an archive of 17,384 official

[7] Children were selected when KLPS-4 began, in 2018 for the first survey wave and in 2020 for the second. For each respondent, at most one child was selected from among their biological children aged 2.5 to 5.5 years at that time and at most one from among those aged 5.5 to 8.5 years (Hsu et al., 2026).

[8] Specifically, the assessments include the Peabody Picture Vocabulary Test (PPVT, Dunn and Dunn, 2007), an animal naming test, the Uwezo Kiswahili, English, and math tests (Africa, 2021), forward and backward digit span, and Raven's Progressive Matrices.

KCPE records that Ozier (2018) collected from District Education Offices and, where those offices lacked records, directly from primary schools. The archive covers the years in which most KLPS cohort members would have taken the KCPE (1999–2006) and records each candidate's name, sex, exam year, primary school, and total score. Because some paper records were lost, the archive covers 88% of the records sought (Ozier, 2018).

Respondents are matched to these records using an updated procedure that compares every name a respondent has reported across KLPS rounds with the candidate's name, requires agreement on primary school, and checks that the sex and exam year are consistent with the respondent's survey data (see Appendix B for details on the approach). Only matches identified automatically by this procedure are accepted. When a respondent matches records from more than one exam year, the analysis uses the earliest score consistent with the respondent's schooling history, which for students who retook the exam is their first attempt. The procedure yields 3,259 matches. After dropping 58 matches with implausible exam years and 39 records with missing scores, the analysis sample includes 3,162 respondents (1,397 women and 1,765 men), about 1,000 more than those successfully matched in earlier work (Ozier, 2018).

Appendix Table A1 compares the analysis sample with the full KLPS cohort. Because it includes only respondents who completed primary school, took the KCPE, and could be matched to an official exam record, the analysis sample is more educated on average than the full cohort (72% of those matched attended secondary school, compared with 53% overall) and includes a smaller share of women, who tended to graduate from primary school at somewhat lower rates in that time period.

Appendix Figure A2 plots the distribution of KCPE scores by gender in the full archive and in the matched sample, and shows that the distribution of scores in the matched sample closely tracks the full archive. The secondary school entrance cutoffs described in Section 2 fall near the median of these distributions, so many students scored close to them. Specifically, the female cutoff of 234 is at about the 45th percentile of female scores in the archive

(and 52nd in the matched sample), and the male cutoff of 251 is at about the 42nd percentile of male scores (48th in the matched sample). Thus the "local" treatment effects estimated near these cutoffs are relevant for a population of individuals with scores close to the average among rural western Kenyan students during this time period.

4 Empirical Strategy

4.1 Regression Discontinuity Design

This paper uses a regression discontinuity (RD) design to estimate the long-run effects of scoring above the KCPE cutoff for admission to government secondary school. Students who completed secondary school had higher exam scores and came from more advantaged backgrounds than those who did not, so differences in their later outcomes cannot necessarily be attributed to schooling alone. Near the admission cutoff, however, small differences in exam scores generated large differences in access to secondary school. The design exploits this discontinuity by comparing students who scored just above and just below the cutoff.

Following Ozier (2018), the admission cutoffs are identified separately by gender using the structural break approach of Card et al. (2008). For each gender, secondary school completion is regressed on the KCPE score, allowing for a discontinuity at a candidate cutoff and for different slopes on either side, and the candidate cutoff that maximizes the R^2 is selected. This procedure yields 234 points for women and 251 points for men, the same cutoffs used by Ozier (2018).[9] Repeating the search separately for groups of exam years produces less precise estimates and does not identify clear alternative cutoffs (Appendix Figure A3), so a single cutoff for each gender is used throughout.

Let K_i denote respondent i's first KCPE score minus the cutoff for the respondent's

[9] Appendix Figure A1 presents the structural break estimates. The male cutoff is one point above 250, the score widely regarded as a passing grade (Section 2). Men who scored exactly 250 completed Form 4 at about the same rate as men who scored just below it, so a cutoff of 250 yields a smaller first stage for men (11 percentage points, not statistically significant).

gender, and let $D_i = \mathbb{1}\{K_i \geq 0\}$ indicate scoring at or above the cutoff. The estimating equation is a local linear regression of the form

$$Y_{it} = \alpha + \tau D_i + \beta_1 K_i + \beta_2 (K_i \times D_i) + \gamma F_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where Y_{it} is an outcome for respondent i in survey round t, F_i is an indicator for female, and λ_t denotes survey-round fixed effects, included only for outcomes observed in multiple rounds. Because the pooled sample combines two gender-specific cutoffs, F_i plays the role of the cutoff fixed effects used in RD designs that pool several cutoffs (Pop-Eleches and Urquiola, 2013; Beuermann and Jackson, 2022); it is omitted from the separate regressions run for women and men. The coefficient τ measures the discontinuity in the outcome at the admission cutoff. The identifying assumption is that, in the absence of the admission rule, potential outcomes would vary smoothly with KCPE scores at the cutoff. Under this assumption, τ identifies the causal effect of scoring above the admission threshold for students with scores close to it.

Estimation uses a triangular kernel and the mean squared error (MSE) optimal bandwidth for each outcome, selected by the Stata `rdrobust` command (Calonico et al., 2014, 2019), and the tables report conventional local linear point estimates with robust bias-corrected standard errors and p-values.[10] Estimates using bandwidths from half to twice the MSE-optimal bandwidth are similar, although several are smaller at the widest bandwidths, and the earnings estimate (which is close to zero throughout) changes sign at twice the optimal bandwidth (Appendix Figure A4). For outcomes observed in several rounds, observations are pooled and error terms are clustered by respondent. Estimates are reported for the pooled sample, with scores centered at the gender-specific cutoffs, as well as separately for women and men.

Because scoring above the cutoff increased the probability of secondary schooling without

[10] Although KCPE scores were originally recorded as integers, rescaling scores from 1999–2000 and centering at gender-specific cutoffs yields more than 160 distinct score values within each bandwidth used in the pooled analysis and more than 80 within each gender-specific bandwidth.

guaranteeing admission or completion, the design is a fuzzy RD. While the main estimates are reduced-form effects of crossing the cutoff rather than instrumental variables (IV) estimates of the effect of completing secondary school, for the cognitive outcomes fuzzy RD estimates that instrument for secondary school completion with the cutoff are also reported, following Ozier (2018) (Panel C of Table 2). Formally, an IV interpretation would require that scoring above the cutoff affects later outcomes only through secondary school completion. Yet admission could also potentially change students' expectations, peer groups, or parental investments, even among students who do not complete secondary school (although this is hard to directly test). Because crossing the cutoff changes secondary completion for only about one in six students, IV estimates are more than five times larger than the corresponding reduced-form estimates and have relatively wide confidence intervals.

The estimates apply to students near the admission thresholds, which as noted above fall near the medians of the gender-specific KCPE score distributions (Section 3). They capture the effects of crossing the admission cutoff for cohorts who took the KCPE before the 2008 tuition reform, rather than the effects of secondary schooling for all students.

4.2 Validity of the RD Design

This subsection assesses the validity of the RD design by examining the distribution of KCPE scores, balance in predetermined characteristics, matching to administrative records, and survey participation.

If students could precisely manipulate their scores around the admission thresholds (for instance, through some forms of cheating), the density of scores would change at the cutoff. Density tests find no evidence of such a discontinuity, either in the pooled sample or separately for women and men (Appendix Figure A5). Reassuringly, predetermined characteristics, including year of birth, grade in 1998, parents' education, assignment to the deworming program, and baseline school characteristics, are also balanced at the cutoff for both women and men (Appendix Table A2). In the pooled sample, the share of women is

6.8 percentage points lower just above the cutoff ($p = 0.06$). However, the pooled estimates control for gender, and allowing the relationship between outcomes and the KCPE scores to differ by gender leave them nearly unchanged (for example, 0.177 rather than 0.175 for Form 4 completion).

A further possible concern is that the analysis includes only respondents whose KLPS records could be linked to official KCPE scores. If the probability of a successful match changes at the cutoff, the matched sample could differ systematically on either side of it. In the full KCPE archive, the probability that a record is matched to a KLPS respondent does not change at the cutoff for men or in the pooled sample, but it falls by 5 to 6 percentage points for women (Appendix Table A4). This drop does not appear to change the composition of the analysis sample. Among matched women, the density of scores and predetermined characteristics are smooth at the cutoff. The drop is also concentrated within a few points of the cutoff, consistent with unmatched second attempts by students who narrowly missed the cutoff and retook the exam (Appendix C).

Finally, survey participation is not significantly different for respondents just above and just below the cutoff in most rounds (Appendix Table A3). The probability of being interviewed in KLPS-2 or later, and in KLPS-4 or KLPS-5, does not change at the cutoff for either women or men. Participation in individual rounds sometimes differs by one gender (KLPS-2 for men and KLPS-5 for women), but the share of respondents observed in the later rounds is similar on both sides of the cutoff. The exceptions are two KLPS-5 modules. Respondents above the cutoff are 8 percentage points less likely to have completed the C+ cognitive module, mainly among men, and men above the cutoff are 13 percentage points less likely to have a child assessed. Section 5 returns to these differences when discussing the cognitive and intergenerational results.

5 Results

5.1 Educational Attainment

Table 1 reports the effects of scoring above the test score cutoff – and thus gaining admission to public secondary schools – on educational attainment. Respondents just above the cutoff are 17.5 percentage points (standard error 5.4) more likely to complete Form 4, the final year of secondary school, than respondents just below it. This represents a nearly 60% increase relative to the below-cutoff completion rate of 30 percent. The effect is similar to the 15 percentage point increase estimated by Ozier (2018) in the original sample; differences between that work and this estimate are due to a greater follow-up time frame (allowing more people to complete their schooling) as well as the larger analysis sample due to improved matching, noted above. Crossing the cutoff also raises the probability of attending any secondary school by 15.7 percentage points and increases years of schooling attained by 0.98 years. Figure 1 graphically presents the jump in completion at the cutoff.

The increase in measured schooling is concentrated in secondary school. Figure 2 shows that the probability of completing at least 9, 10, 11, and 12 years of schooling – corresponding to Forms 1 through 4 in secondary school – rises by 16 to 18 percentage points at the cutoff. The similar effects on entry and completion suggest that most students induced to enter secondary school by crossing the test score cutoff remain enrolled through Form 4. The probability of continuing beyond secondary school also rises meaningfully by 8.9 percentage points (denoted by 13/14 years of schooling attainment in the figure), significant at the 10% level, but there is no detectable effect on university completion (denoted by 15 to 16 years of schooling attainment).

These results suggest that scoring just above the test score cutoff changes whether students attend secondary school at all, rather than only which school they attend. Among respondents just below the cutoff, 31% never attend secondary school. This distinguishes the setting from many RD studies of school admission, in which students on both sides of the

cutoff attend secondary school but enter schools of different selectivity or quality (Jackson, 2010; Pop-Eleches and Urquiola, 2013; Abdulkadiroğlu et al., 2014; Lucas and Mbiti, 2014; Beuermann and Jackson, 2022; Cabrera-Hernández et al., 2026). These studies estimate the effects of attending a better school, whereas the effect of attending any secondary school at all may be much larger, as Ozier (2018) notes. Here, scoring a few points above the cutoff increases the probability of completing secondary school and adds nearly one year of schooling attainment on average.

The increase in secondary school completion is similar for women and men (Appendix Table A5). Women just above the cutoff are 22.0 percentage points more likely to complete Form 4 and complete 1.06 more years of schooling, while men are 21.0 percentage points more likely to complete Form 4 and complete 0.90 more years.[11] Neither difference across genders is statistically significant ($p = 0.90$ for Form 4 completion and $p = 0.88$ for years of schooling).[12] Men above the cutoff are also 13.1 percentage points more likely to continue beyond secondary school, significant at the 10% level, while the corresponding estimate for women is somewhat smaller at 5.1 percentage points and not statistically significant.[13]

5.2 Cognition

Table 2 reports the effects on adult cognition, measured in KLPS-5 roughly two decades after most respondents took the KCPE. Respondents just above the cutoff score 0.24 standard deviations higher on the general cognition factor than respondents just below it (Panel B).

[11] The pooled estimate (17.5 percentage points) is smaller than either gender-specific estimate because the MSE-optimal bandwidth is wider in the pooled sample: 43 points, compared with 30 for women and 34 for men. At the pooled bandwidth, the estimates are 18.0 percentage points for women and 16.8 for men.

[12] These tests of differences between women and men use a local linear regression in which every term is interacted with an indicator for women, estimated with the pooled MSE-optimal bandwidth, a triangular kernel, and heteroskedasticity-robust standard errors.

[13] Earlier work finds that the long-run labor market effects of deworming in this cohort are concentrated among men and among respondents older than 12 at the 1998 baseline (Hamory et al., 2021). Appendix Table A6 reports the effects on attainment separately by baseline cohort and gender. The point estimates are generally larger for the older cohort, but none of the differences between cohorts is statistically significant (p-values range from 0.37 to 0.99) and these age differences are not emphasized in what follows.

Among the five domains, the largest effect is on memory, at 0.26 standard deviations.[14] Both estimates are significant at the 5% level. The estimate for visuospatial ability is 0.21 standard deviations and significant at the 10% level, while the estimates for orientation (0.19), executive function (0.14), and language (−0.01) are not statistically significant. Orientation and visuospatial ability show strong ceiling effects in this sample, because the HCAP was designed for adults aged 60 and older, so estimates for these domains are less informative (Gross et al., 2025). For instance, the orientation questions ask respondents where they currently are and what day of the week it is, which nearly all middle-aged individuals are able to answer correctly.

These effects appear quite large relative to the increase in schooling at the cutoff. Recall that scoring above the cutoff adds approximately one year of schooling on average, implying a gain of about 0.25 standard deviations in general cognition per additional year of schooling, if the effect operates entirely through higher schooling attainment. Panel C reports fuzzy RD estimates that instrument for secondary school completion with the cutoff, following Ozier (2018).[15] If the exclusion restriction discussed in Section 4 holds, the estimated effect of completing secondary school is 1.25 standard deviations for general cognition and 1.47 standard deviations for memory. These estimates are relatively imprecise, however, with a 95% confidence interval of 0.05 to 2.46 standard deviations for general cognition.

These findings extend the results of Ozier (2018), who estimated that completing secondary school raised a combined vocabulary and reasoning score by 0.6 standard deviations when respondents were in their early 20s. Nearly two decades after the exam, respondents above the cutoff continue to perform substantially better on a broader battery of cognitive assessments, with the largest gains in memory. There are many mechanisms that could

[14]The relatively large effect on memory is consistent with evidence from compulsory schooling reforms in high-income countries, which find lasting improvements in memory decades after schooling (Glymour et al., 2008; Schneeweis et al., 2014; Gorman, 2023). In a Scottish cohort tested at age 11 and again at age 70, education predicts scores at age 70 most strongly on Logical Memory, a memory test also used in this study, even after controlling for intelligence at age 11(Ritchie et al., 2015).

[15]Panel A reports OLS estimates comparing respondents who did and did not complete secondary school, which also reflect selection into schooling. Note that the OLS estimate for general cognition is 0.71 standard deviations, within the IV confidence interval.

have contributed to this effect, and they are not mutually exclusive and may reinforce each other. One immediate explanation is that certain cognitive skills acquired in secondary school are retained into adulthood. Another is that schooling changes subsequent experiences in ways that reinforce cognitive skills, for example by increasing access to more cognitively demanding occupations or encouraging greater use of reading, math, computers and other cognitively stimulating activities. More complex work has been associated with higher cognitive functioning (Schooler et al., 1999), and a compulsory schooling reform in England and Wales both increased later-life memory and changed the occupations people entered (Gorman, 2023). These mechanisms may operate together, and occupational differences could reinforce cognitive skills even in the absence of detectable earnings gains.

The estimated impacts on general cognition are similar for women and men, at 0.21 and 0.25 standard deviations (Appendix Table A7). While the pooled estimate is significant at the 5% level (as noted above), the estimate for men alone is significant at the 10% level, while the estimate for women is not statistically significant; the difference between effects for women and men is not statistically significant ($p = 0.89$). The effect on memory is 0.33 standard deviations for men, significant at the 10% level, and 0.19 for women, and this difference is also not statistically significant ($p = 0.54$).

One concern is differential attrition: men just above the cutoff are 11 percentage points less likely to have completed the cognitive module, significant at the 5% level, while women's completion does not change (Appendix Table A3). However, there is limited evidence that this attrition biases the estimates. Among respondents who did not complete the cognitive module, those just above and just below the cutoff had similar Raven's scores in earlier survey rounds (KLPS-2 and KLPS-3). Moreover, Lee bounds (Lee, 2009), which account for differential attrition under worst-case assumptions about the missing respondents, remain positive, ranging from 0.10 to 0.37 standard deviations for general cognition in the pooled sample, although the lower bound is not statistically significant (Appendix Table A8).[16]

[16]Raven's scores in KLPS-2 and KLPS-3 were measured when respondents were approximately 23 years old. Among respondents who did not complete the cognitive module, the discontinuity in these scores is −0.03

These results bear on the concern that expanding schooling in LMICs may produce little actual learning (Pritchett, 2013; World Bank, 2018). For students near the middle of the KCPE score distribution, the implied gain of 0.25 standard deviations in general cognition per additional year of schooling falls within the range of 0.07 to 0.33 standard deviations estimated in a meta-analysis of quasi-experimental studies, mostly from high-income countries (Ritchie and Tucker-Drob, 2018). Causal evidence from LMICs is more limited and has mostly examined academic test scores rather than broader cognitive function later in adulthood. Randomized evaluations of secondary school scholarships in Ghana and conditional cash transfers in Nicaragua find improvements in knowledge and academic skills (Duflo et al., 2025; Barham et al., 2024), while a regression discontinuity study of scholarships in Cambodia finds no improvement in test scores despite increased schooling (Filmer and Schady, 2014). Evidence on cognition later in life is particularly scarce. A natural experiment created by China's Great Famine suggests that completing primary school raised cognitive scores at older ages (Huang and Zhou, 2013), and using partial-identification bounds rather than experimental or quasi-experimental variation, Amin et al. (2025) find gains in memory from primary schooling in four of the five LMICs, but cannot rule out zero effects at higher levels of schooling. The causal estimates here show that access to secondary school at this margin produced cognitive gains that remain large into midlife.

These results may also have implications for cognitive aging. Low educational attainment is seen as a major modifiable risk factor for dementia, and is estimated to account for 7% of dementia cases worldwide (Livingston et al., 2020), and the number of people living with dementia in eastern Sub-Saharan Africa is projected to more than quadruple between 2019 and 2050 (GBD 2019 Dementia Forecasting Collaborators, 2022). Respondents in KLPS were assessed in their mid-30s to early 40s, before substantial age-related cognitive decline typically begins (Gross et al., 2025). Future rounds of the KLPS can examine whether the

standard deviations in the pooled sample and −0.01 among men. Restricting the sample to respondents who completed the module changes the discontinuity in earlier Raven's scores from 0.03 to 0.05 standard deviations in the pooled sample and from 0.25 to 0.34 among men.

cognitive gains found here persist into older ages and whether secondary schooling affects the rate of later cognitive decline, rather than just the level of cognitive functioning.

5.3 Fertility and Family Formation

In the pooled sample, scoring above the cutoff reduces the number of children by 0.29, or 9% of the below-cutoff mean, but the estimate is not statistically significant (Table 3, Panel A). Scoring above the cutoff does not change the probability of having any children or child mortality. However, it does lead to a reduction in fertility among men. By the most recent survey round (KLPS-5), when most respondents were in their late 30s, men just above the cutoff have 0.80 fewer children than men just below it, a 24% decline from a mean of 3.3, significant at the 5% level (Appendix Table A10).[17] Women above the cutoff have 0.22 more children on average, but this effect is not statistically significant and the 95% confidence interval rules out declines larger than 0.56 children (Appendix Table A9). The difference between women and men is statistically significant ($p = 0.03$). This indicates that men's and women's fertility respond differently to secondary schooling in this population, since women and men gained similar amounts of schooling at the cutoff.

The decline in men's fertility emerges gradually over time, well after most men would have left secondary school. Figure 3 plots the effect on cumulative fertility at each age from 15 to 40. For men, the effect is close to zero through age 22, reaches about 0.4 children by age 26, and exceeds one child by age 36 (Panel C). Because the gap opens after men have left school, it cannot solely be explained by the contemporaneous effect of enrollment on childbearing, sometimes called an incarceration effect (Black et al., 2008; Duflo et al., 2026). The reduction is concentrated in the number of children men have rather than whether they become fathers at all: men above the cutoff are just as likely to have any children, but among fathers, the number of live births falls by 1.06. Men's first births also occur about one year

[17] Men may not report, or know about, children born outside marriage or with former partners (Duflo et al., 2026). Such underreporting would bias the RD estimates only if it changed at the cutoff. Men above the cutoff could be more likely to acknowledge children, understating the decline, or less likely, overstating it. It is challenging to establish the magnitude of any such effects.

later, but the estimated effect on timing is not statistically significant.

The timing documented here also helps explain why shorter follow-ups in other studies may miss the decline. In the related Ghana study, men's fertility did not change significantly by about age 30, but men in the comparison group had had fewer than one child on average by that age (Duflo et al., 2026). Because men in the sample can still have more children even as they enter their 40s and beyond, future rounds of the KLPS will allow us to test whether the measured effect persists or simply reflects a delay in childbearing among men above the cutoff.

For women, the estimated effects on fertility are small and negative through age 30 and small and positive from age 32, and none is statistically significant (Figure 3, Panel B). Ozier (2018) had found that secondary school reduced teen pregnancy in this cohort. In the expanded sample, the estimated effect on a first birth before age 18 is also large and negative but imprecisely estimated (−5.6 percentage points, from a base of 6 percent), and the age at first birth rises by a non-significant 0.6 years. These results are consistent with secondary school delaying women's childbearing without reducing the number of children they eventually have by their late 30s on average.[18]

Changes in marriage and partner selection could potentially help explain any changes in fertility: later marriage reduces the years available for childbearing within marriage, while changes in spouses' characteristics or marital stability could also affect family size. Table 4 reports the effects on marriage, separately for women and men. Marriage is nearly universal in this cohort: below the cutoff, 87% of women and 92% of men have ever married by the most recent survey round. Scoring above the cutoff does not significantly change the probability of ever marrying or the number of marriages for either gender. Age at first marriage rises by a meaningful 1.1 years for women and 0.7 years for men, but neither estimated effect is

[18]Once again the contrast with the Ghana study is informative. There scholarships reduced pregnancy among women in their early 20s, but fertility partially caught up by about age 30, and female scholarship recipients were also more likely to have regular salaried employment, consistent with a higher opportunity cost of childbearing (Duflo et al., 2026). The effects on earnings in this sample are discussed below (and in Appendix Table A11).

statistically significant. The age gap with spouses, spouses' education and earnings, and use of modern contraception with a spouse also do not change significantly. The one exception is divorce: men above the cutoff are 7.2 percentage points more likely to have ever divorced, from a base of 6 percent, which is significant at the 5% level. Yet because this is one of 16 marriage estimates (the others of which are not significant), it should be interpreted with caution due to multiple hypothesis testing issues.

The limited changes in marriage outcomes found in Kenya contrast with findings from other settings. In the Ghana study discussed above, female secondary school scholarship recipients were less likely to be married or cohabiting in their late 20s and more likely to have partners with tertiary education, while male recipients were more likely to live with their parents (Duflo et al., 2026). The differences between the settings may partly reflect the longer follow-up in the KLPS, which tracks respondents into their late 30s, by which point nearly everyone has married. The positive point estimates for age at first marriage, together with the later age at first birth among women, in KLPS are also consistent with some delay in family formation, although none of these effects is statistically significant. In a study in Mali, expanded school access reduced both men's probability of having a child by age 25 and the number of children living in their households, with evidence pointing to later marriage and greater urban residence (Godefroy and Lewis, 2018). In the KLPS sample, by contrast, men's fertility falls mainly after age 25, without detectable changes in marriage timing or partner selection.

Schooling could also reduce fertility by changing individuals' preferences for family size, and there is some evidence for this in the Kenya study sample. Scoring above the test score cutoff reduces desired family size, mainly among women: in the pooled sample, respondents above the cutoff want 0.36 fewer children in KLPS-2, 0.25 fewer in KLPS-3, and 0.28 fewer in KLPS-4 (significant at the 5% level in KLPS-2 and at the 10% level in the later rounds, Table 3, Panel B). Women above the cutoff want about 0.4 fewer children in each round, while the estimates for men are smaller and not statistically significant. The pattern of

effects is, perhaps surprisingly, almost reversed for desired versus realized fertility: women report wanting fewer children without a detectable reduction in the number they have, whereas men have substantially fewer children without a statistically significant change in their stated desired family size.

One possibility for this pattern is that schooling changes fertility decisions within partnerships rather than primarily delaying marriage. In most LMICs, men are thought to want to have more children than women, and when women have more bargaining power, a couple's fertility tends toward the lower of the two spouses' preferences (Doepke and Tertilt, 2018). If schooling leads men to give more weight to their partners' preferences, couples could end up having fewer children even without a change in men's own desired family size. The data cannot directly test this explanation, and there is no detectable effect on modern contraceptive use with a spouse (which might be related to the spouse's preferences). Changes in marital stability could also contribute, given the increase in divorce among men. In all, the results thus suggest that changes in marriage timing and partner selection are unlikely to account for the full decline in men's fertility, but that other factors may play a role.

5.4 Earnings and Consumption

This section reports the effects on earnings and consumption, pooling the survey rounds in which each outcome was measured (Appendix Table A11). Despite the observed gains in cognition noted above, scoring above the test score cutoff has no detectable effect on total earnings or consumption. The estimated effect on annual non-farm earnings, measured from respondents' early 20s to their late 30s (across multiple survey rounds), is −$87, relative to a below-cutoff mean of $2,197 (2024 USD PPP). This effect is relatively imprecisely estimated and is not significant at traditional levels: the 95% confidence interval ranges from a decline of 36% to an increase of 22 percent. Because the cutoff raises schooling by about one year, the upper end of this interval corresponds to a return of about 22% per year of schooling if the effect operates entirely through schooling, so the estimates cannot rule out substantial

returns. The estimated effect on per capita household consumption, measured in KLPS-3 and KLPS-4, is −$63, relative to a mean of $3,610, and again is not statistically significant.

Disaggregating earnings by source, there are no significant effects on wage earnings, self-employment profits, or household earnings. Farming profits fall by $30, significant at the 5% level, but average only $30 per year for those below the cutoff, so the decline is small in absolute terms.[19] Ozier (2018) found that secondary schooling reduced low-skill self-employment among men in their early 20s. In the longer-run data, men's self-employment profits are also slightly lower above the cutoff, but the difference is small and not significant.

One possible explanation for the lack of detectable earnings gains is that returns to schooling are concentrated in formal wage jobs, which are relatively scarce in LMIC settings like Kenya. Evidence from urban Ghana and Tanzania shows that returns to education are high in large firms but lower among the self-employed (Rankin et al., 2010). Wage employment is relatively limited in the KLPS, particularly among women: in KLPS-5, about half of men and only a third of women report any wage employment (Byambaa et al., 2025). In Ghana, secondary school scholarships increased women's earnings over a decade later, partly via public-sector jobs, such as teaching and nursing, that are typically reached through tertiary training (Duflo et al., 2025). As discussed above, students above the test score cutoff do not have higher university completion overall, and women in particular are not more likely to continue beyond secondary school (Appendix Table A5). The cognitive gains documented in this Kenya sample may therefore have limited effects on earnings if secondary schooling does not also improve access to jobs that reward these skills.

5.5 Children's Outcomes

Table 5 reports the effects on children aged 8 to 14, separately by the gender of the parent (from the KLPS sample). Children of women who are just above the cutoff score 0.36 standard deviations higher on the academic index than children of women just below it,

[19] Estimating separately by gender, the only significant estimate is a $37 decline in men's farming profits, significant at the 10% level.

with the largest gain in language (0.41 standard deviations). Both estimates are significant at the 10% level. The estimates for math, overall cognition, and executive function are also positive but not statistically significant, and the estimate for fluid intelligence is close to zero. Among the children of KLPS men, the estimates for the academic, language, and math indices are negative and not statistically significant. The difference between the effects on the academic index for children of women and men, 0.54 standard deviations, is significant at the 10% level, providing suggestive evidence that there are larger gains among children of women than men who are just above the threshold.[20] Among the other indices, children's behavior, health, and schooling investment do not change for the pooled sample or for the children of either women or men. There is a statistically significant gain (of 0.34 standard deviation units) in the home environment index for children of men; this index includes measures of books, magazines, and pictures in the home, the variety of play materials, and play activities.

Because children were assessed only at ages 8 to 14, changes in fertility or the timing of births could potentially affect the composition of the sample. However, the cognitive gains for children of women do not appear to reflect changes in which children were assessed: the probability that at least one child was assessed does not change at the cutoff for women, although it falls by 13 percentage points for men, significant at the 10% level (Appendix Table A3). For both, the assessed children are similar on either side of the cutoff in sex, age, and whether they were the older or younger child selected (Appendix Table A12), although mothers above the cutoff were about one year older when the child was born, consistent with somewhat delayed childbearing. Controlling for children's characteristics raises the estimate for the academic index to 0.41 standard deviations, and also controlling for mother's age at birth yields 0.37 standard deviations, while the estimate for the children of men remains about -0.18 standard deviations (Appendix Table A13).

[20]The regression interacts every term of the local linear regression with an indicator for female respondents, as in the tests for the education and cognition outcomes. The p-value is 0.045 with heteroskedasticity-robust standard errors and 0.064 with standard errors clustered by respondent.

The gains for children of women are similar to findings from Ghana, where secondary school scholarships for women raised their children's cognitive scores by about 0.25 standard deviations at ages 5 and 7, while scholarships for men had no detectable effects (Duflo et al., 2026). The KLPS results extend this pattern to older children (aged 8 to 14 years old) and to academic skills measured during primary school. Studies of schooling reforms in Norway and Sweden also find effects on children's outcomes only from mothers' schooling (Black et al., 2005; Lundborg et al., 2014). In the KLPS, women and men gain similar amounts of schooling at the cutoff, as well as similar gains in general cognition (at 0.21 and 0.25 standard deviations, respectively). The larger effects on children of women, therefore, do not appear to simply reflect larger human capital gains among mothers. Instead, the point estimates suggest that the effects of improved parent human capital depend on parent gender.

Several channels could explain the gains for children of women, including greater household resources, smaller families, and increased investments in children's learning, but the estimates provide little direct evidence for these channels. Among women scoring above the cutoff, the number of children, their spouse's schooling attainment, and schooling investment in the child do not change significantly. While per capita household earnings rise by \$433, or 29% of the below-cutoff mean, like other living standards estimates this estimate is imprecise and not statistically significant. The home environment index – which measures learning materials and activities in the home – also does not change significantly for the children of women; ironically, it rises for the children of men, whose academic skills do not appear to improve. In the Ghana study, the gains for children appear to operate mainly through caregivers' nonmonetary investments, such as cognitive stimulation (Duflo et al., 2026), which the home environment index in KLPS may not fully capture.

6 Conclusion

Governments across Sub-Saharan Africa are expanding access to secondary school, and about half of the countries in the region now offer fee-free public secondary school (Garlick, 2019; Ferreira and Sandholtz, 2026). Yet evidence on the long-run effects of secondary schooling in LMICs remains limited. This paper uses the KCPE admission cutoff in Kenya and a panel that follows students for nearly two decades after the exam to estimate effects on cognition, fertility, economic outcomes, and outcomes of children in the next generation.

Scoring just above the cutoff raises secondary school completion by 18 percentage points, with similar increases for women and men. Nearly two decades after the exam, respondents above the cutoff score 0.24 standard deviations higher on a general cognition factor, with the largest gain in memory. By their late 30s, men above the cutoff have about 0.8 fewer children, while women's fertility does not change meaningfully. Despite the substantial cognitive gains, there are no detectable increases in earnings or household consumption, although these estimates are relatively imprecise. There is also suggestive evidence of gains in academic skills among the children of women who scored above the cutoff, but not among the children of men.

These results have three implications. First, evaluations of secondary schooling that rely only on measured earnings would miss its effects on cognitive skills. At this margin, secondary schooling in Kenya produced cognitive gains that lasted into midlife, despite widespread concern that expanding schooling in LMICs produces little learning (Pritchett, 2013; World Bank, 2018). The lack of detectable earnings gains suggests that improvements in cognitive skills may not directly translate into higher earnings without access to formal jobs that reward them, although the estimates cannot rule out substantial earnings returns, reflecting well-known concerns about the measurement of living standards in low- and middle-income country settings.

Second, the long-run effects of schooling on fertility can differ markedly from those measured in early adulthood. Women's early fertility response documented in Ozier (2018) did

not lead to fewer children by their late 30s, while the decline in men's fertility emerged only in their mid-20s and widened through their 30s. Studies that focus on women and end in early adulthood may therefore miss important effects on men's fertility while mistakenly interpreting delays in women's childbearing as reductions in fertility.

Third, the gains from secondary schooling may extend beyond the individuals getting more schooling themselves. Because women and men gained similar amounts of schooling at the test score cutoff in this population, the larger gains for the children of women are consistent with mothers' schooling having greater effects on children's learning (rather than simply reflecting differences in parent human capital). If these estimates reflect causal effects, evaluations that count only students' own outcomes understate the returns to secondary schooling for girls, which matters for decisions about public funding of free secondary education (Duflo et al., 2026).

These RD findings apply to students near the test score admission cutoffs and to cohorts who took the KCPE before Kenya abolished secondary school tuition fees in 2008. Expansions that reach entire cohorts at once may have different effects, for example, if they change who attends secondary school or the value of a secondary school credential in the labor market as a whole. Because the estimates here capture the effects of crossing the admission cutoff, they may also reflect changes that admission produced apart from additional schooling, such as different peers or expectations, although these are hard to quantify.

There remain several open questions. Future survey rounds can determine whether men's lower fertility persists through middle age (when some men still have children), as well as whether the cognitive gains extend into older ages or affect the rate of later cognitive decline.

Table 1: Education Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Years of Schooling	10.10	0.979***	(0.319)	30.1	1,368	3,103
Any Secondary Schooling	0.57	0.157***	(0.052)	34.5	1,528	3,103
Completed Form 4	0.30	0.175***	(0.054)	43.0	1,865	3,103
13+ Years of Schooling	0.16	0.089*	(0.055)	34.7	1,533	3,103

Notes: Educational attainment is measured from harmonized annual schooling histories in KLPS rounds 1–4 (1998–2021). *Years of Schooling* is the highest grade attained, with vocational training counted only for respondents who had completed Form 4; *Any Secondary Schooling*, *Completed Form 4*, and *13+ Years of Schooling* indicate attaining at least 9, 12, and 13 years. Attainment is defined only for respondents interviewed in KLPS-2 (2007–2009) or later; those last observed earlier were typically still enrolled. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 2: Cognitive Impacts

Outcome	Comparison mean	Estimate	(SE)	Band-width	Eff. obs.	Obs.
Panel A: Secondary Completers versus Non-Completers (OLS)						
General Cognitive Factor Score	0.03	0.712***	(0.034)	–	–	2,418
Memory Factor Score	-0.03	0.595***	(0.038)	–	–	2,418
Language Factor Score	0.04	0.553***	(0.035)	–	–	2,418
Executive Function Factor Score	0.08	0.632***	(0.034)	–	–	2,418
Orientation Factor Score	0.06	0.162***	(0.035)	–	–	2,418
Visuospatial Factor Score	0.11	0.363***	(0.035)	–	–	2,418
Panel B: Reduced Form (Intent to Treat)						
General Cognitive Factor Score	0.05	0.241**	(0.123)	31.0	1,122	2,451
Memory Factor Score	-0.02	0.264**	(0.131)	34.4	1,224	2,451
Language Factor Score	0.08	-0.013	(0.141)	29.4	1,063	2,451
Executive Function Factor Score	0.09	0.139	(0.126)	29.9	1,067	2,451
Orientation Factor Score	0.06	0.187	(0.135)	32.6	1,158	2,451
Visuospatial Factor Score	0.13	0.210*	(0.134)	28.4	1,015	2,451
Panel C: Fuzzy RD (IV), Completing Secondary School						
General Cognitive Factor Score	0.05	1.249**	(0.615)	37.5	1,311	2,418
Memory Factor Score	-0.02	1.474**	(0.710)	38.8	1,350	2,418
Language Factor Score	0.08	-0.002	(0.599)	34.4	1,209	2,418
Executive Function Factor Score	0.09	0.662	(0.601)	42.9	1,455	2,418
Orientation Factor Score	0.06	0.889	(0.686)	32.8	1,144	2,418
Visuospatial Factor Score	0.13	0.870	(0.686)	42.4	1,454	2,418

Notes: Cognitive outcomes are domain factor scores from the KLPS-5 C+ Module, which fielded the Harmonized Cognitive Assessment Protocol in Kenya (Gross et al., 2025): one score per domain, estimated by confirmatory factor analysis from a battery of more than 20 assessments and standardized to mean zero and unit variance in the full KLPS-5 sample, so means in the analysis sample need not equal zero. Panel A reports OLS coefficients on an indicator for completing secondary school, with heteroskedasticity-robust standard errors; the bandwidth and effective-observation columns do not apply (–). Panel B reports reduced-form RD estimates of scoring at or above the admission cutoff. Panel C reports fuzzy RD (IV) estimates in which completing secondary school is instrumented by an indicator for scoring at or above the cutoff, with the same kernel and bandwidth selection. *Comparison mean* is the mean among non-completers in Panel A and among respondents scoring below the cutoff in Panels B and C. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), and RD estimates in Panels B and C control for an indicator for female respondents (Section 4). RD estimates are conventional local-linear coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points and *Eff. obs.* the observations within it. Standard errors (in parentheses) and significance stars in Panels B and C are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 3: Fertility Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Panel A: Realized Fertility						
Indicator for any children	0.89	0.043	(0.047)	32.0	1,462	3,154
Total number of children	3.23	-0.292	(0.250)	34.0	1,548	3,154
Age at first birth	23.71	0.834	(0.608)	35.2	1,363	2,669
Number of live births (cond> 0)	3.64	-0.508**	(0.235)	32.2	1,246	2,669
Under five mortality	0.05	0.014	(0.021)	34.9	3,562	7,242
Under one mortality	0.03	0.002	(0.017)	34.5	4,350	8,862
Panel B: Fertility Intentions						
Desired No. Children (KLPS2)	3.37	-0.363**	(0.196)	27.4	1,024	2,550
Desired No. Children (KLPS3)	3.32	-0.252*	(0.141)	40.2	1,464	2,562
Desired No. Children (KLPS4)	4.06	-0.282*	(0.178)	40.1	1,411	2,449

Notes: Panel A reports realized fertility from retrospective birth histories in KLPS rounds 1–5: an indicator for any children, children ever born, age at first birth, live births among respondents with at least one child, and under-five and under-one mortality. The mortality outcomes are measured at the pregnancy level; their standard errors are clustered by respondent and their observation counts are pregnancies. Panel B reports the respondent's stated ideal number of children, including those already born, from KLPS-2, KLPS-3, and KLPS-4. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 4: Marriage Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
Ever married	0.87	0.009	(0.061)	27.4	547	1,344
Number of marriages	1.02	0.030	(0.097)	26.4	531	1,344
Age at marriage	22.10	1.066	(0.831)	34.9	587	1,143
Age gap with spouse	-5.41	0.946	(0.836)	24.3	428	1,138
Education level of spouse	10.18	0.192	(0.731)	34.6	580	1,131
Spouse's earnings	376.62	88.409	(94.892)	30.8	513	1,116
Modern birth control with spouse	0.66	-0.051	(0.099)	36.1	576	1,065
Ever divorced	0.08	-0.033	(0.058)	34.7	584	1,139
Among males:						
Ever married	0.92	0.014	(0.057)	33.9	799	1,699
Number of marriages	1.33	-0.071	(0.148)	29.6	706	1,699
Age at marriage	25.00	0.665	(0.850)	31.1	660	1,477
Age gap with spouse	4.04	0.530	(0.595)	39.3	812	1,475
Education level of spouse	9.48	-0.326	(0.541)	32.1	677	1,470
Spouse's earnings	159.03	-65.232	(60.592)	43.5	862	1,449
Modern birth control with spouse	0.55	0.044	(0.095)	35.1	683	1,363
Ever divorced	0.06	0.072**	(0.038)	36.5	754	1,476

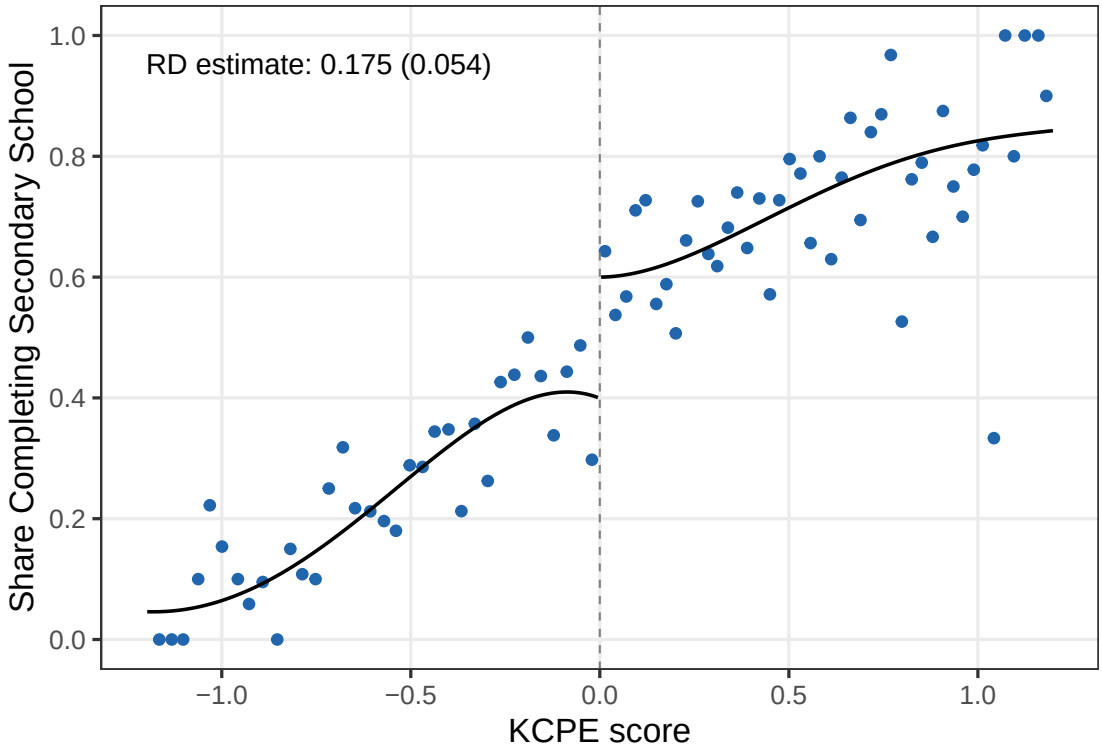
Notes: Marriage outcomes are constructed from the KLPS marriage histories. Age at marriage, age gap with spouse (the respondent's age at marriage minus the spouse's), spouse's education and earnings, and modern birth control with spouse describe the respondent's *first* marriage; ever married, number of marriages, and ever divorced cover *all* marriages. Spouse's earnings are winsorized and expressed in 2024 USD PPP. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table 5: Intergenerational Impacts by Parent's Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Children of female respondents:						
Overall cognitive index	0.15	0.307	(0.215)	34.2	465	943
Academic index	0.16	0.362*	(0.212)	33.3	452	943
Language index	0.17	0.414*	(0.243)	31.0	410	943
Math index	0.13	0.245	(0.173)	42.2	581	943
Executive function index	0.10	0.215	(0.231)	35.4	492	943
Fluid intelligence index	0.09	-0.033	(0.276)	43.5	309	491
Strengths & difficulties index	0.11	0.105	(0.179)	38.6	533	945
Subjective health index	0.04	0.065	(0.195)	39.8	546	943
Home environment index	0.06	-0.044	(0.229)	32.6	433	941
Schooling investment index	0.09	0.129	(0.142)	31.4	423	945
Children of male respondents:						
Overall cognitive index	0.04	-0.012	(0.256)	31.0	503	1,102
Academic index	0.04	-0.172	(0.247)	30.5	488	1,102
Language index	0.06	-0.086	(0.225)	35.5	577	1,102
Math index	0.02	-0.208	(0.259)	27.8	447	1,102
Executive function index	0.04	0.259	(0.240)	32.8	524	1,102
Fluid intelligence index	-0.01	-0.074	(0.382)	34.3	276	556
Strengths & difficulties index	0.03	-0.143	(0.253)	38.5	614	1,105
Subjective health index	0.04	-0.164	(0.248)	30.1	484	1,100
Home environment index	0.06	0.336**	(0.196)	29.8	478	1,104
Schooling investment index	-0.01	-0.097	(0.220)	31.2	503	1,106

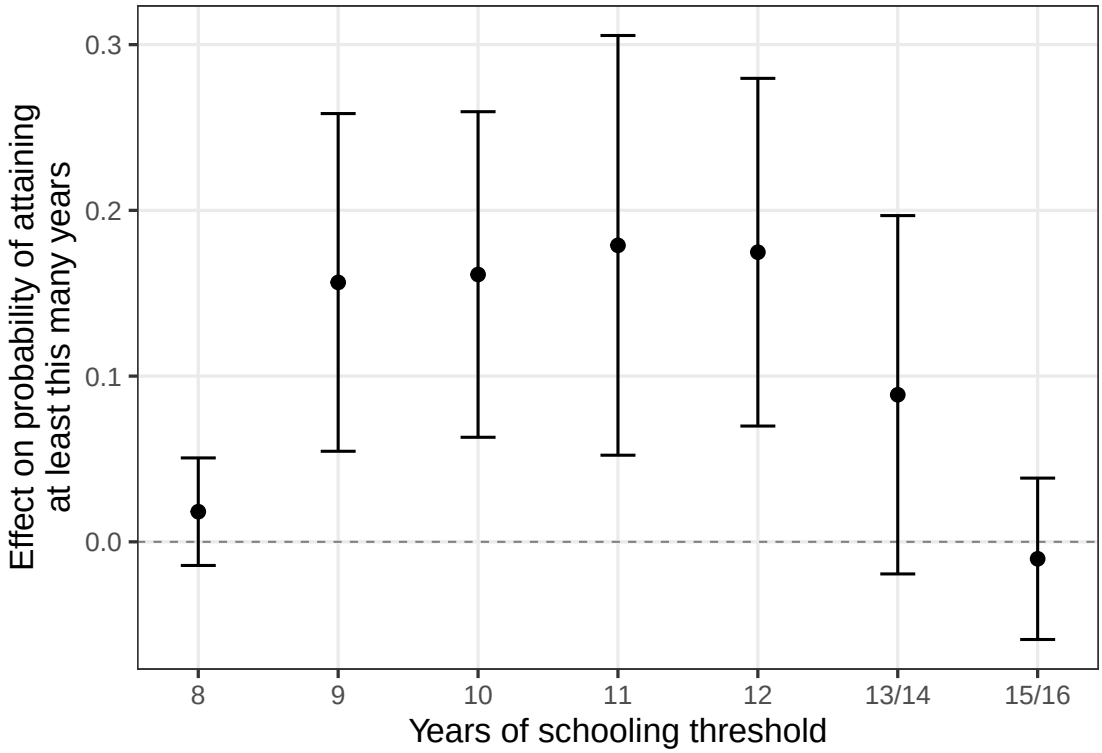
Notes: Outcomes are summary indices from the KLPS-5 child module, administered to up to two biological children aged 8–14 of each respondent; observations are children. The *Language* index combines the PPVT, animal naming, and Uwezo Swahili and English assessments; *Math* the Uwezo math assessment; *Executive function* digit span; and *Fluid intelligence* Raven's matrices, administered only to older children. The *Academic* index combines Language and Math, and the *Overall cognitive* index combines Language, Math, and Executive function. The *Strengths and difficulties* index combines the five SDQ subscales, with difficulties reverse-coded so that higher values are better; *Subjective health*, *Home environment*, and *Schooling investment* combine caregiver-reported items on the child's health, learning materials and activities in the home, and school enrollment, attendance, and spending. Each index is standardized to mean zero and unit variance in the full KLPS-5 child sample. Panels split the sample by the *parent's* sex, not the child's, with KCPE scores re-centered at the parent's gender-specific cutoff (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among children of parents scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Figure 1: Secondary School Completion



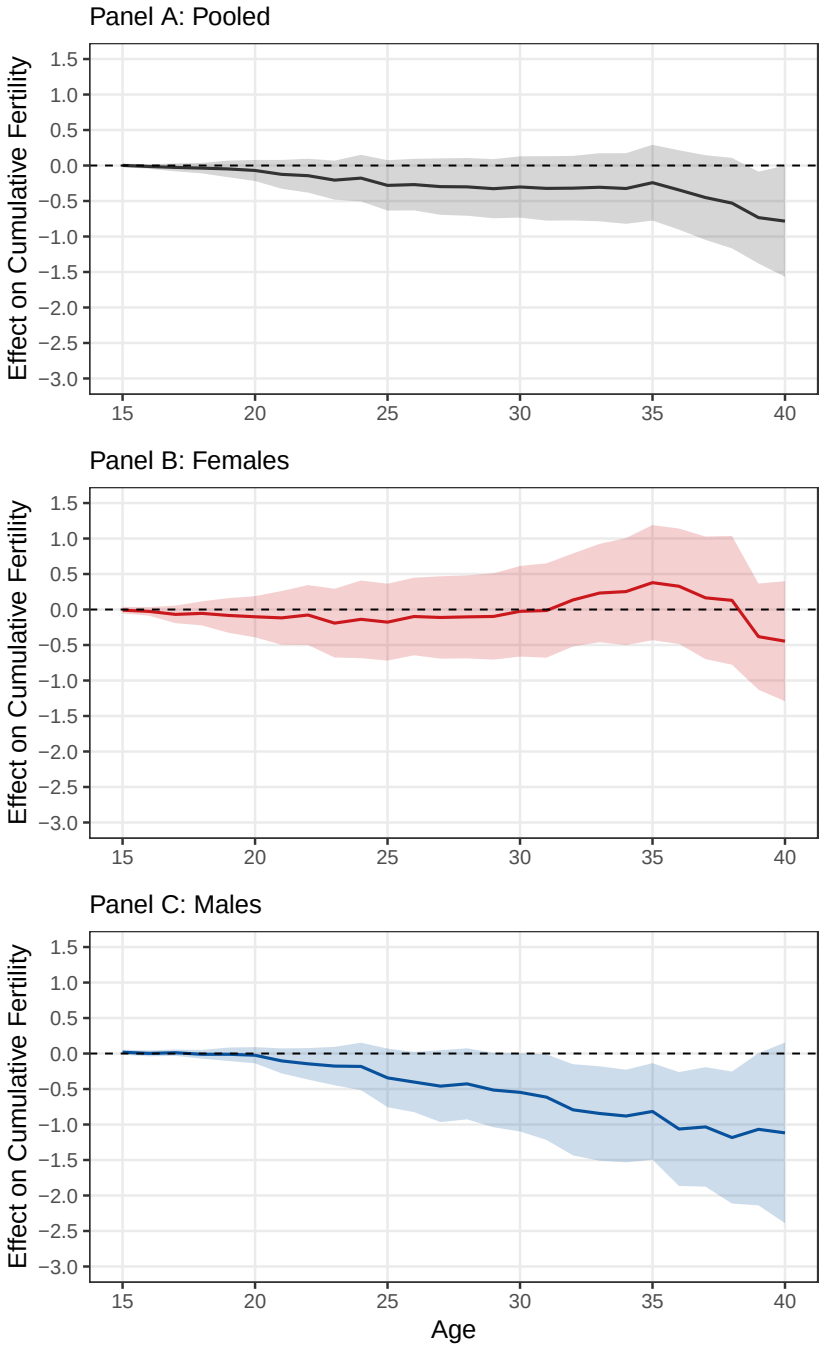
Notes: The figure plots the probability of completing Form 4 (the final year of secondary school) against the KCPE score, re-centered so that zero is the gender-specific cutoff. Completion is constructed from retrospective schooling histories in KLPS rounds 1–4 (see Table 1 for the education variables). Points are mean completion within evenly spaced `rdplot` bins of the running variable; the solid lines are local-linear fits estimated separately on each side of the cutoff. The printed estimate is the conventional local-linear RD coefficient at the cutoff, controlling for an indicator for female respondents, with its robust bias-corrected standard error in parentheses, as in Table 1.

Figure 2: Effects on Years-of-Schooling Thresholds



Notes: Each point is a separate conventional local-linear RD estimate, with its robust bias-corrected 95 percent confidence interval, of the effect of scoring at or above the cutoff on the probability of attaining at least the number of years of schooling on the horizontal axis, controlling for an indicator for female respondents. Years of schooling are the harmonized KLPS attainment measures, in the same sample as Table 1. The effect at 8 years is mechanically near zero, since everyone in the sample reached standard 8 to sit the KCPE. Attainment is recorded at 12, 14, and 16 years for completed secondary, some tertiary, and university, so no respondent attains exactly 13 or 15 years; the 13- and 14-year thresholds therefore define the same indicator, as do the 15- and 16-year thresholds, and each pair is shown as a single estimate (13/14 and 15/16).

Figure 3: Effect on Cumulative Fertility by Age



Notes: Each panel plots reduced-form RD estimates of the effect of scoring above the KCPE cutoff on cumulative fertility (children ever born by each age from 15 to 40), from retrospective birth histories in KLPS rounds 1–5. Panel A pools men and women, with scores re-centered at the gender-specific cutoff and a control for an indicator for female respondents; Panels B and C estimate separately for women and men (cutoffs of 234 and 251). Age is the calendar year minus the respondent's year of birth at baseline. The solid line is the conventional local-linear estimate age by age; shaded bands are 95 percent confidence intervals from the robust bias-corrected standard errors of `rdrobust` with MSE-optimal bandwidths. Negative values indicate fewer births among those just above the cutoff. Each age uses the respondents whose birth history extends to that age; roughly three-quarters of the sample contributes at age 36 and one-third at age 40, since the youngest cohorts have not yet reached the oldest ages.

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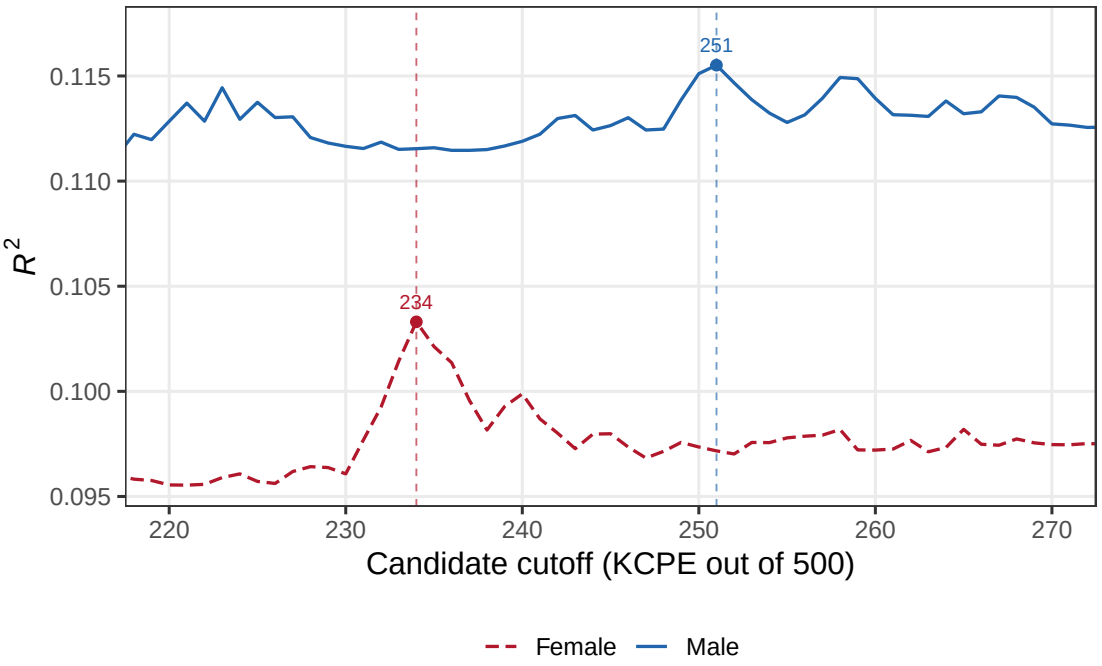
A Additional Tables and Figures

Table A1: Summary Statistics: KLPS Cohort and Matched Analysis Sample

	Full KLPS cohort	Matched analysis sample
Baseline characteristics		
Female	0.51 (0.50)	0.44 (0.50)
Year of birth	1985.4 (2.2)	1985.4 (2.2)
Age in 2024	38.6 (2.2)	38.6 (2.2)
Grade in 1998	4.48 (1.70)	4.78 (1.65)
Deworming treatment (early phase-in)	0.66 (0.47)	0.66 (0.47)
KCPE		
KCPE score (500-point scale)	–	244.6 (51.3)
Scored at or above cutoff	–	0.51 (0.50)
Survey participation		
Interviewed in KLPS-2 or later	0.88 (0.33)	0.98 (0.14)
Interviewed in KLPS-5 (to date)	0.71 (0.45)	0.77 (0.42)
Age at KLPS-5 interview	39.1 (2.3)	39.1 (2.2)
Adult outcomes		
Years of schooling	9.67 (3.06)	11.15 (2.63)
Any secondary schooling	0.53 (0.50)	0.72 (0.45)
Completed Form 4	0.32 (0.47)	0.50 (0.50)
Ever married	0.87 (0.33)	0.86 (0.35)
Number of marriages	1.15 (0.70)	1.10 (0.67)
Any children	0.86 (0.35)	0.85 (0.36)
Number of children	3.15 (2.16)	2.87 (2.00)
Lives in urban area (KLPS-4)	0.57 (0.49)	0.64 (0.48)
Total non-farm earnings, KLPS-5 (2024 USD PPP)	2,826 (4,061)	3,743 (4,770)
Any non-farm earnings, KLPS-5	0.78 (0.41)	0.81 (0.39)
Respondents	7,527	3,155

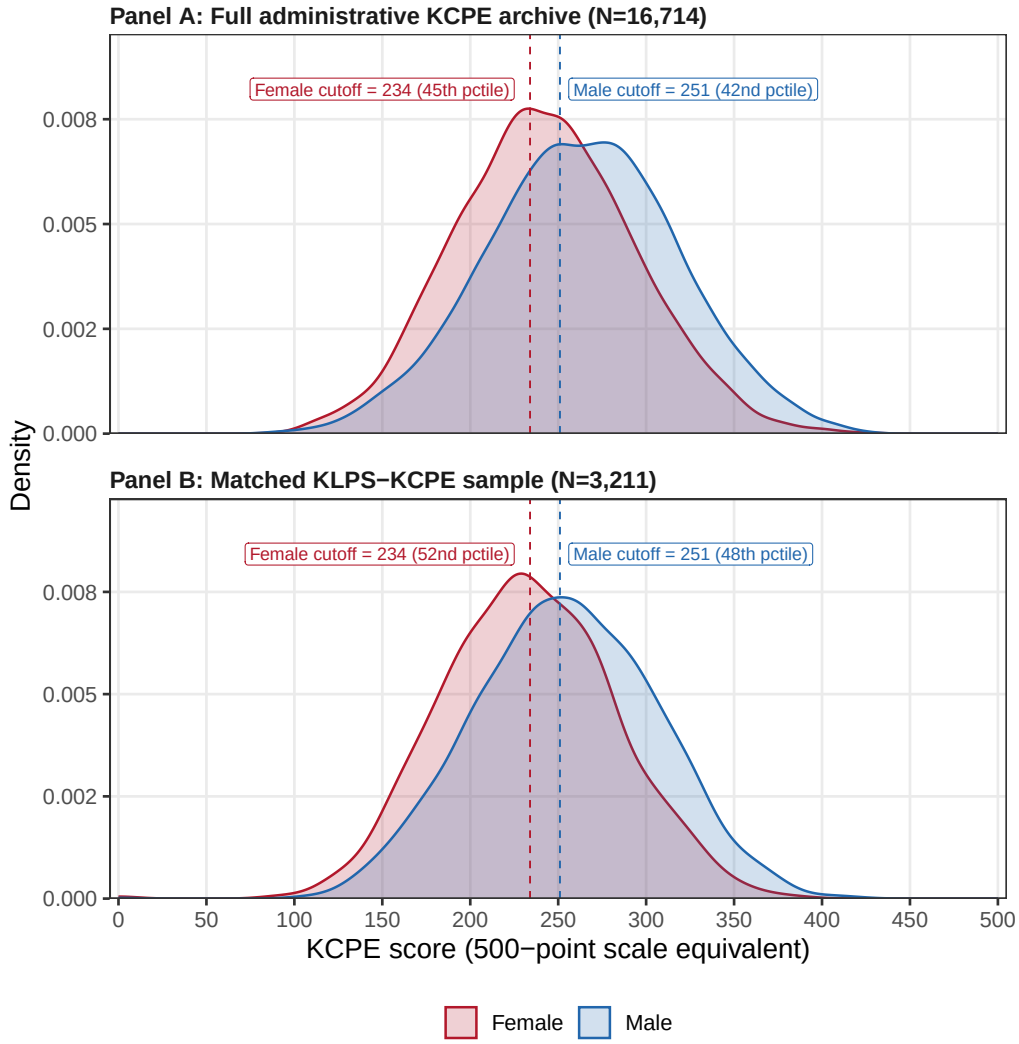
Notes: Means, with standard deviations in parentheses. The first column is the original KLPS cohort—pupils enrolled in grades 2–7 in the 75 Primary School Deworming Program schools in 1998; the second is the subset linked to a KCPE record that enters the analysis sample (Appendix B). *Female*, *Year of birth*, *Grade in 1998*, and *Deworming treatment* come from the 1998 baseline roster; year of birth is missing for about one-fifth of the cohort, and *Age in 2024* is 2024 minus year of birth. Linkage to a KCPE record requires having sat the examination, which accounts for the higher attainment, urbanization, and earnings of the matched sample. *KCPE score* is the first-attempt total on the 500-point scale (scores from the 1999–2000 examinations converted from 700 points), and *Scored at or above cutoff* refers to the gender-specific cutoffs of 234 for females and 251 for males. *Age at KLPS-5 interview* is the interview year minus year of birth among respondents interviewed to date. Schooling outcomes are defined for respondents interviewed in KLPS-2 or later, as in Table 1; marriage and fertility outcomes follow Tables 4 and 3; *Lives in urban area* is measured at the KLPS-4 interview, the most recent round with a harmonized urban indicator; earnings are from completed KLPS-5 interviews, in 2024 USD PPP, trimmed of the top one percent. Each row uses the respondents for whom the variable is observed.

Figure A1: Structural Break Search



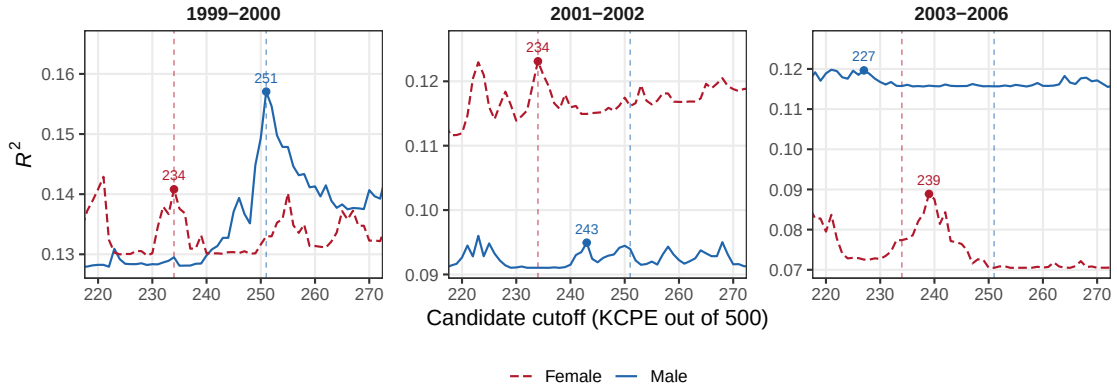
Notes: The figure implements a structural-break search in the spirit of Card et al. (2008) for the KCPE score at which the probability of completing secondary school (Form 4) rises discontinuously. For each candidate cutoff between 200 and 300 on the 500-point scale we fit a piecewise-linear model for completion within a symmetric ±70-point window and plot its fit (R^2), separately for females and males; scores from the pre-2001 700-point examinations are converted to 500-point equivalents. The figure displays the 220–270 range, where the fits peak; the R^2 profiles decline outside it. The female series peaks at 234 and the male series at 251, marked by the vertical lines; these are the cutoffs used throughout, as in Ozier (2018). Appendix Figure A3 repeats the search separately by exam-year group.

Figure A2: KCPE Score Distributions by Gender



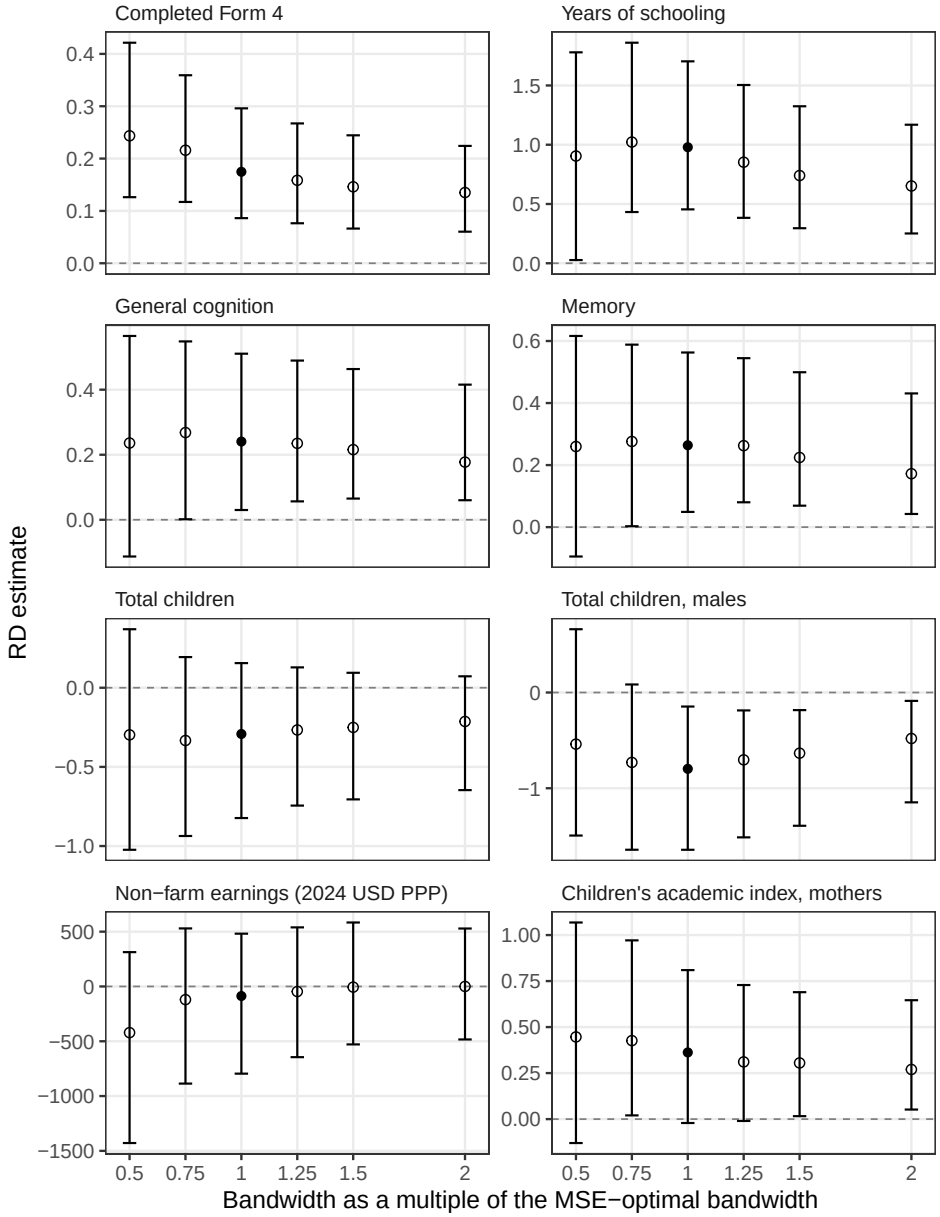
Notes: The figure plots kernel densities of KCPE scores for female and male examinees, in the full administrative KCPE archive (Panel A) and in the matched KLPS–KCPE sample (Panel B). Scores from the 1999–2000 examinations, graded out of 700 points, are converted to 500-point equivalents, so all scores are shown on the 500-point scale. Panel A includes the 16,714 archive records from examination years 1999–2006 with a valid score and recorded sex (of 17,384 records in the archive overall); Panel B includes the 3,211 matched KLPS respondents with a verified KCPE score and exam year in the same range, before the cohort and exam-timing restrictions that define the analysis sample. Dashed vertical lines mark the gender-specific admission cutoffs (234 for females, 251 for males), and each label reports the percentile of the relevant gender-specific score distribution at which that cutoff falls. Both cutoffs fall near the middle of the distribution, where the data are dense, and the matched sample closely tracks the archive from which it is drawn.

Figure A3: Structural Break Search by KCPE Year Group



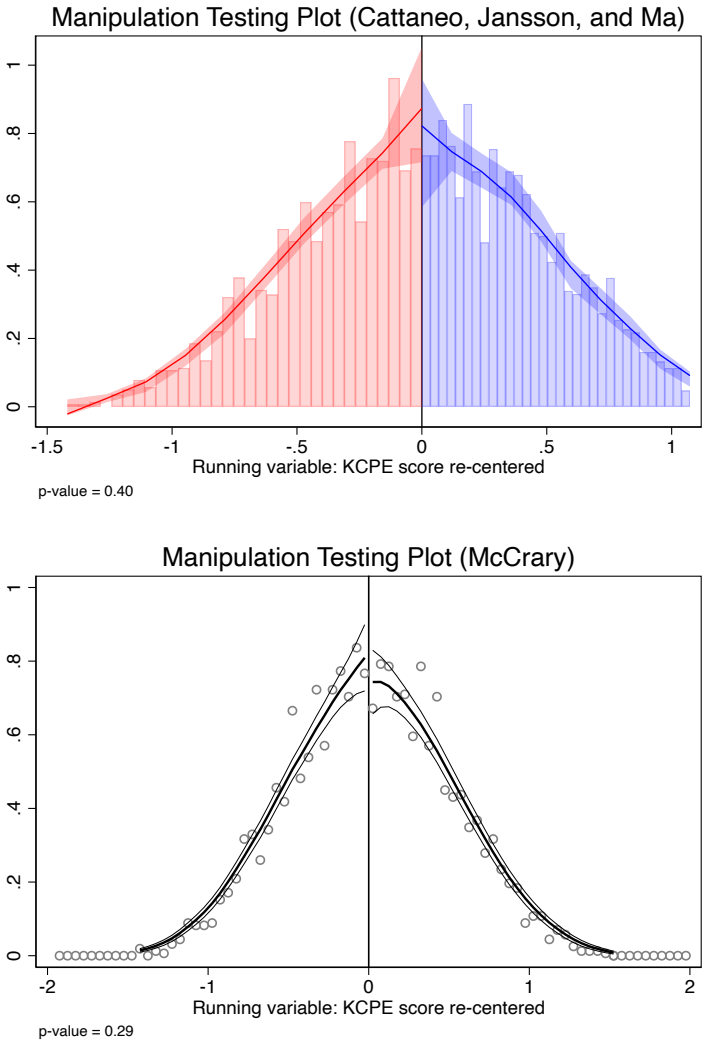
Notes: This figure repeats the structural-break search of Appendix Figure A1 separately by KCPE examination year group (1999–2000, 2001–2002, and 2003–2006) and by gender. For each candidate cutoff between 200 and 300 on the 500-point scale, we fit the same piecewise-linear model for Form 4 completion within a symmetric ±70-point window in each gender-by-year-group cell and plot the resulting R^2; as in Appendix Figure A1, the figure displays the 220–270 range. Scores from 1999–2000 are converted to 500-point equivalents, and the 2003–2006 group folds in the small number of examinees who sat the KCPE in 2006. Dashed vertical lines mark the pooled gender-specific cutoffs (234 for females, 251 for males); dots mark each cell's R^2-maximizing candidate between 225 and 275 (among candidates implying a positive discontinuity). Each cell contains roughly 280–620 observations within the search window, versus roughly 1,100–1,400 pooled, so the subperiod profiles are flatter and noisier. The female search peaks at exactly 234 in both 1999–2000 and 2001–2002 and at 239 in 2003–2006; the male search peaks at 251 in 1999–2000, matching the pooled search, while the flatter 2001–2002 and 2003–2006 male profiles peak at 243 and 227. In three of the six cells the full-range maximizer lies outside this interior range (at 213, 223, and 298), but each sits in a nearly flat profile, exceeding the interior peak by at most 0.012: those subsamples carry little information about the precise threshold location, in contrast with the sharply peaked pooled search.

Figure A4: Sensitivity of the Main Estimates to the Bandwidth



Notes: Each point is a conventional local-linear RD estimate using a bandwidth equal to the indicated multiple of the outcome's MSE-optimal bandwidth, with the bias-correction bandwidth scaled by the same multiple. Bars are robust bias-corrected 95 percent confidence intervals (Calonico et al., 2014). Filled points use the MSE-optimal bandwidth and reproduce the estimates in Tables 1, 2, 3, and 5 and Appendix Tables A10 and A11. The MSE-optimal bandwidths range from 29 to 43 KCPE points. Pooled estimates control for an indicator for female respondents. Non-farm earnings pool KLPS-2 through KLPS-5, include survey-round indicators, and cluster standard errors by respondent. The children's academic index is for children of female respondents.

Figure A5: Density test



Notes: The figure reports manipulation tests for the running variable—the KCPE score in the matched analysis sample (N=3,162), re-centered at the gender-specific cutoff (234 for females, 251 for males) and pooled across genders. The top panel is the local-polynomial density test of Cattaneo et al. (2020) (`rddensity`); the bottom panel is the McCrary (2008) discontinuity test (`DCdensity`). Both fail to reject continuity of the density at the cutoff (p-values reported in each panel), consistent with no sorting in the test score around the threshold. Local-polynomial density tests run separately by gender also fail to reject ($p = 0.92$ for females and $p = 0.77$ for males), and Appendix Table A4 reports an archive-side test of whether record linkage is smooth through the cutoff.

Table A2: Covariate Balance at the KCPE Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Female	0.46	-0.068*	(0.042)	32.8	1,480	3,162
Year of birth	1984.89	0.148	(0.272)	29.1	1,267	2,938
Grade in 1998	4.98	0.095	(0.165)	39.5	1,766	3,162
Father's education	9.31	0.313	(0.739)	34.5	1,451	2,940
Mother's education	6.04	0.663	(0.523)	35.9	1,511	2,966
Deworming Treatment	0.66	-0.022	(0.054)	42.1	1,864	3,155
School average test score, 1996	-0.02	0.021	(0.073)	39.6	1,763	3,155
Pupils at baseline school, 1998	398.37	16.656	(56.794)	42.3	1,864	3,155
Among females:						
Year of birth	1985.19	-0.024	(0.380)	30.4	594	1,292
Grade in 1998	5.03	0.253	(0.269)	33.4	690	1,397
Father's education	9.75	-0.025	(1.128)	30.4	585	1,289
Mother's education	6.19	0.574	(0.771)	34.2	662	1,305
Deworming Treatment	0.67	0.004	(0.089)	29.9	618	1,390
School average test score, 1996	-0.03	-0.050	(0.085)	29.9	618	1,390
Pupils at baseline school, 1998	396.21	53.022	(58.474)	35.6	733	1,390
Among males:						
Year of birth	1984.62	0.277	(0.405)	30.7	710	1,646
Grade in 1998	4.94	-0.074	(0.275)	34.0	834	1,765
Father's education	8.93	0.357	(1.134)	30.7	711	1,651
Mother's education	5.91	0.781	(0.810)	31.3	727	1,661
Deworming Treatment	0.66	0.003	(0.075)	44.5	1,051	1,765
School average test score, 1996	-0.01	0.073	(0.088)	49.3	1,149	1,765
Pupils at baseline school, 1998	400.23	-7.473	(61.281)	43.1	1,031	1,765

Notes: The table reports RD estimates for characteristics determined before the KCPE examination, in the matched analysis sample. *Female*, *Year of birth*, and *Grade in 1998* come from the 1998 baseline pupil roster. *Father's education* and *Mother's education* are the highest levels attained, in years, as reported in KLPS-2 (as in Ozier 2018), filled where missing from the corresponding KLPS-3 and then KLPS-4 questions converted to the same years scale; together these cover about 93 percent of the matched sample. *Deworming Treatment* indicates assignment to the first two phase-in groups (treated from 1998–99) of the Primary School Deworming Program (Miguel and Kremer, 2004). *School average test score, 1996* (in standard-deviation units) and *Pupils at baseline school, 1998* describe the baseline primary school; these two rows cluster standard errors by school. The *Female* row appears only in the pooled panel, since pooling stacks the two gender-specific cutoffs. The other pooled rows control for an indicator for female respondents (Section 4). KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A3: Survey Participation at the KCPE Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Interviewed in KLPS-2 or later	0.98	-0.020	(0.022)	36.6	1,647	3,162
Interviewed in KLPS-2	0.82	-0.097**	(0.048)	30.2	1,393	3,162
Interviewed in KLPS-3	0.83	-0.009	(0.051)	36.8	1,651	3,162
Interviewed in KLPS-4	0.77	0.010	(0.048)	42.7	1,872	3,162
Interviewed in KLPS-5 (to date)	0.90	-0.038	(0.035)	41.9	1,841	3,162
Interviewed in KLPS-4 or KLPS-5	0.94	-0.028	(0.028)	34.9	1,562	3,162
Completed C+ cognitive module	0.82	-0.078*	(0.048)	37.9	1,689	3,162
Any child assessed, ages 8–14	0.56	-0.048	(0.061)	35.9	1,614	3,162
Among females:						
Interviewed in KLPS-2 or later	0.98	-0.019	(0.026)	36.9	761	1,397
Interviewed in KLPS-2	0.82	-0.060	(0.069)	34.0	706	1,397
Interviewed in KLPS-3	0.81	0.022	(0.085)	32.3	665	1,397
Interviewed in KLPS-4	0.79	0.079	(0.073)	33.4	690	1,397
Interviewed in KLPS-5 (to date)	0.93	-0.078*	(0.056)	30.6	638	1,397
Interviewed in KLPS-4 or KLPS-5	0.95	0.000	(0.040)	29.7	620	1,397
Completed C+ cognitive module	0.84	-0.020	(0.073)	30.4	638	1,397
Any child assessed, ages 8–14	0.57	0.059	(0.099)	28.5	588	1,397
Among males:						
Interviewed in KLPS-2 or later	0.98	-0.023	(0.037)	34.5	849	1,765
Interviewed in KLPS-2	0.82	-0.167**	(0.084)	20.0	547	1,765
Interviewed in KLPS-3	0.86	-0.025	(0.058)	46.6	1,090	1,765
Interviewed in KLPS-4	0.75	-0.031	(0.067)	42.8	1,011	1,765
Interviewed in KLPS-5 (to date)	0.88	-0.015	(0.054)	35.4	873	1,765
Interviewed in KLPS-4 or KLPS-5	0.93	-0.054	(0.044)	32.2	804	1,765
Completed C+ cognitive module	0.81	-0.113**	(0.064)	43.1	1,031	1,765
Any child assessed, ages 8–14	0.55	-0.132*	(0.090)	31.5	782	1,765

Notes: The table reports RD estimates for survey participation in the matched analysis sample. *Interviewed in KLPS-2 or later* is the measurement rule for educational attainment (Table 1). The round-specific rows indicate completion of a main interview in that round; KLPS-5 is ongoing, so its row reflects interviews completed to date, and *Interviewed in KLPS-4 or KLPS-5* combines the two most recent rounds. *Completed C+ cognitive module* indicates completion of the KLPS-5 cognitive assessment (Table 2); *Any child assessed, ages 8–14* indicates that at least one biological child of the respondent completed the child assessment module. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Pooled estimates control for an indicator for female respondents (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A4: KCPE Archive Match Rates at the Cutoff

Outcome	Below-cutoff mean	RD estimate		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
All examinees:						
Matched to a KLPS respondent	0.22	-0.019	(0.021)	38.9	8,694	16,714
In the analysis sample	0.22	-0.016	(0.021)	38.1	8,659	16,714
Female examinees:						
Matched to a KLPS respondent	0.23	-0.056**	(0.032)	36.7	3,718	7,153
In the analysis sample	0.23	-0.049*	(0.032)	36.0	3,621	7,153
Male examinees:						
Matched to a KLPS respondent	0.22	0.011	(0.028)	41.5	5,110	9,561
In the analysis sample	0.21	0.010	(0.028)	40.7	5,011	9,561

Notes: The table reports RD estimates, over KCPE archive records, of the probability that a record is linked to the KLPS—a test of whether linkage is smooth through the cutoff. The sample is the 16,714 archive records from examination years 1999–2006 with a valid score and recorded sex; scores from 1999–2000 are converted to 500-point equivalents, and the panels split records by the sex recorded in the archive. *Matched to a KLPS respondent* indicates linkage by the procedure of Appendix B; *In the analysis sample* additionally requires that the linked respondent enters the analysis sample: of 3,259 accepted matches, 58 are dropped because the exam year falls more than one year before or three years after the year implied by the respondent's baseline grade and 39 because the matched record lacks a recorded score, leaving 3,162. Match rates are low because the archive also contains cohorts outside the KLPS sampling frame and repeat attempts: at most 4,948 records—one per KLPS respondent who reached standard 8—could link, so the maximum attainable rate is 0.30, and the realized below-cutoff rate is about three-quarters of it. The female match rate falls by 5–6 percentage points at the cutoff; Section 4 and Appendix C present evidence that this reflects repeat examination-taking rather than differential linkage of respondents, and that the density of matched respondents and covariate balance are smooth at the cutoff for both genders (Appendix Figure A5; Appendix Table A2). KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates for all examinees control for an indicator for female examinees (Section 4). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among records scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A5: Education Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
Years of Schooling	9.78	1.061**	(0.491)	28.5	577	1,369
Any Secondary Schooling	0.49	0.243***	(0.097)	24.7	521	1,369
Completed Form 4	0.25	0.220***	(0.094)	30.4	627	1,369
13+ Years of Schooling	0.13	0.051	(0.077)	36.6	742	1,369
Among males:						
Years of Schooling	10.38	0.895**	(0.434)	32.5	792	1,734
Any Secondary Schooling	0.64	0.105*	(0.063)	41.3	970	1,734
Completed Form 4	0.35	0.210***	(0.086)	33.7	818	1,734
13+ Years of Schooling	0.18	0.131*	(0.085)	30.5	746	1,734

Notes: The table re-estimates the Table 1 education outcomes separately for females and males; see Table 1 for construction. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A6: Education Impacts by Baseline Cohort and Sex

Outcome	Older cohort (age >12 in 1998)		Younger cohort (age ≤12 in 1998)		Difference	
	Below-cutoff mean	Estimate	Below-cutoff mean	Estimate	Older − younger	*p*-value
Panel A: Full Sample						
Years of Schooling	9.82	1.006***	10.58	0.763	0.243	0.74
		(0.398)		(0.627)	(0.743)	
Any Secondary Schooling	0.51	0.146**	0.66	0.148	-0.002	0.99
		(0.070)		(0.107)	(0.128)	
Completed Form 4	0.25	0.268***	0.38	0.168*	0.100	0.49
		(0.088)		(0.113)	(0.143)	
13+ Years of Schooling	0.12	0.137**	0.22	0.029	0.108	0.39
		(0.078)		(0.100)	(0.126)	
Observations	1,517		1,365			
Panel B: Females						
Years of Schooling	9.46	1.557**	10.21	1.035	0.522	0.65
		(0.708)		(0.895)	(1.141)	
Any Secondary Schooling	0.43	0.225**	0.58	0.246*	-0.021	0.92
		(0.117)		(0.167)	(0.204)	
Completed Form 4	0.19	0.399***	0.32	0.202	0.197	0.37
		(0.157)		(0.154)	(0.220)	
13+ Years of Schooling	0.10	0.126	0.18	0.034	0.092	0.61
		(0.136)		(0.117)	(0.179)	
Observations	626		641			
Panel C: Males						
Years of Schooling	10.10	0.774	10.99	0.413	0.361	0.70
		(0.499)		(0.800)	(0.942)	
Any Secondary Schooling	0.58	0.105	0.75	0.058	0.047	0.73
		(0.083)		(0.107)	(0.136)	
Completed Form 4	0.30	0.207**	0.45	0.103	0.104	0.58
		(0.105)		(0.154)	(0.187)	
13+ Years of Schooling	0.14	0.158*	0.26	0.033	0.125	0.50
		(0.100)		(0.156)	(0.185)	
Observations	891		724			

Notes: The table re-estimates the Table 1 education outcomes separately by baseline cohort and sex; see Table 1 for construction. Cohorts follow Hamory et al. (2021): respondents over age 12 at the 1998 baseline (older) versus age 12 or under (younger), with age and sex from the baseline sample master; 221 respondents lacking a baseline birth year are dropped. Each cell is a separate conventional local-linear RD estimate from `rdrobust` with a triangular kernel and its own MSE-optimal bandwidth, KCPE scores re-centered at the gender-specific cutoffs; Panel A estimates control for an indicator for female respondents (Section 4); robust bias-corrected standard errors are in parentheses, and *Below-cutoff mean* is the outcome mean among below-cutoff respondents in the relevant subsample. *Older − younger* is the difference in the two cohort estimates, with standard error $\sqrt{\text{se}^2_{\text{older}} + \text{se}^2_{\text{younger}}}$ since the subsamples are disjoint; the *p*-value tests equality. Panels B and C partition Panel A. * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A7: Cognitive Impacts by Gender

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Among females:						
General Cognitive Factor Score	-0.03	0.210	(0.161)	38.6	636	1,108
Memory Factor Score	-0.02	0.189	(0.191)	33.7	547	1,108
Language Factor Score	0.03	-0.105	(0.201)	31.8	520	1,108
Executive Function Factor Score	-0.06	0.241	(0.191)	30.2	505	1,108
Orientation Factor Score	-0.01	0.146	(0.201)	38.7	636	1,108
Visuospatial Factor Score	0.00	0.261	(0.191)	32.4	528	1,108
Among males:						
General Cognitive Factor Score	0.12	0.250*	(0.166)	34.3	662	1,343
Memory Factor Score	-0.02	0.332*	(0.202)	32.4	624	1,343
Language Factor Score	0.12	0.068	(0.180)	34.2	662	1,343
Executive Function Factor Score	0.22	0.029	(0.166)	28.6	547	1,343
Orientation Factor Score	0.12	0.246	(0.175)	30.9	587	1,343
Visuospatial Factor Score	0.25	0.126	(0.176)	30.0	571	1,343

Notes: The table re-estimates the reduced-form (Panel B) cognitive outcomes of Table 2 separately for females and males; see Table 2 for the factor-score construction. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males), so the coefficient on the discontinuity may be read directly. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A8: Bounds on Cognitive Impacts Accounting for Attrition

	Reduced form		Bounds				
Outcome	Estimate	(SE)	Lower	Upper	Share trimmed	Band-width	Obs.
All respondents:							
General Cognitive Factor Score	0.241**	(0.123)	0.089	0.366**	0.095	31.0	2,451
Memory Factor Score	0.264**	(0.131)	0.074	0.453***	0.095	34.4	2,451
Among females:							
General Cognitive Factor Score	0.210	(0.161)	0.166	0.223	0.024	38.6	1,108
Memory Factor Score	0.189	(0.191)	0.153	0.216	0.024	33.7	1,108
Among males:							
General Cognitive Factor Score	0.250*	(0.166)	0.033	0.446**	0.136	34.3	1,343
Memory Factor Score	0.332*	(0.202)	0.074	0.602***	0.136	32.4	1,343

Notes: The table reports bounds on the reduced-form cognitive estimates in Table 2 and Appendix Table A7 that account for differential completion of the KLPS-5 cognitive module at the cutoff, following Lee (2009). Completion rates just below and just above the cutoff are estimated by local linear regression with a triangular kernel at the MSE-optimal bandwidth for completion; *Share trimmed* is the difference between the two rates divided by the higher rate. Within the MSE-optimal bandwidth for each outcome, this share of respondents is trimmed from the side of the cutoff with the higher completion rate, from either the top or the bottom of the distribution of residuals from a local linear regression of the outcome on the KCPE score on that side, and the discontinuity is re-estimated at the same bandwidth; the smaller and larger of the two estimates are the lower and upper bounds. *Estimate* is the untrimmed estimate. Pooled estimates control for an indicator for female respondents. Estimates are conventional local-linear RD coefficients from `rdrobust`; standard errors (in parentheses) and significance stars, including those on the bounds, are based on the robust bias-corrected inference of Calonico et al. (2014). *Bandwidth* is the MSE-optimal bandwidth in KCPE points, and *Obs.* is the number of respondents who completed the module. * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A9: Fertility Impacts: Females

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Realized Fertility:						
Indicator for any children	0.89	0.075	(0.072)	29.5	617	1,395
Total number of children	3.14	0.216	(0.395)	29.0	591	1,395
Age at first birth	22.37	0.591	(0.952)	32.0	560	1,200
Teenage pregnancy	0.06	-0.056	(0.051)	31.4	651	1,395
Number of live births (cond> 0)	3.53	-0.038	(0.364)	28.7	503	1,200
Under five mortality	0.06	0.015	(0.035)	29.9	1,418	3,332
Under one mortality	0.04	-0.004	(0.028)	33.5	1,869	3,892
Fertility Intentions:						
Desired No. Children (KLPS2)	3.26	-0.443*	(0.301)	26.4	457	1,125
Desired No. Children (KLPS3)	3.24	-0.390**	(0.205)	47.6	752	1,096
Desired No. Children (KLPS4)	3.85	-0.404*	(0.283)	28.4	471	1,097

Notes: The table re-estimates the Table 3 fertility outcomes for women, adding an indicator for teenage pregnancy (a first birth before age 18, zero for respondents with no children); see Table 3 for construction. Mortality outcomes are measured at the pregnancy level, with standard errors clustered by respondent. KCPE scores are re-centered at the female cutoff of 234. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A10: Fertility Impacts: Males

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Realized Fertility:						
Indicator for any children	0.88	0.023	(0.067)	30.5	755	1,759
Total number of children	3.30	-0.797**	(0.382)	28.8	706	1,759
Age at first birth	24.89	1.038	(0.809)	34.7	718	1,469
Teenage pregnancy	0.03	0.013	(0.020)	30.8	755	1,759
Number of live births (cond> 0)	3.74	-1.059***	(0.360)	27.0	560	1,469
Under five mortality	0.05	0.008	(0.025)	41.5	2,223	3,910
Under one mortality	0.02	0.007	(0.019)	39.8	2,761	4,970
Fertility Intentions:						
Desired No. Children (KLPS2)	3.47	-0.283	(0.235)	35.1	701	1,425
Desired No. Children (KLPS3)	3.39	-0.093	(0.204)	35.1	725	1,466
Desired No. Children (KLPS4)	4.24	-0.348	(0.314)	29.4	570	1,352

Notes: The table re-estimates the Table 3 fertility outcomes for men, adding an indicator for teenage pregnancy (a first birth before age 18, zero for respondents with no children); see Table 3 for construction. Mortality outcomes are measured at the pregnancy level, with standard errors clustered by respondent. KCPE scores are re-centered at the male cutoff of 251. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A11: Earnings and Consumption Impacts

Outcome	Below-cutoff mean	Reduced form		Band-width	Eff. obs.	Obs.
		Estimate	(SE)			
Pooled:						
Total non-farm earnings, 12m	2197.17	-87.275	(325.650)	30.8	4,388	9,761
Farming profits, 12m	30.17	-29.750**	(16.509)	28.2	2,062	4,914
Self-employment profits, 12m	532.84	-26.360	(107.149)	34.6	4,899	9,814
Wage earnings, 12m	1516.28	84.742	(277.411)	34.3	4,873	9,745
Per capita household consumption, 12m	3610.31	-63.371	(488.757)	29.0	1,187	2,687
Total household earnings, 12m	1915.23	249.524	(343.815)	29.1	2,051	4,672
Among females:						
Total non-farm earnings, 12m	1151.05	13.876	(370.089)	29.6	1,983	4,364
Farming profits, 12m	25.73	-16.855	(18.379)	35.3	1,155	2,154
Self-employment profits, 12m	321.04	-4.769	(111.876)	36.6	2,403	4,369
Wage earnings, 12m	753.60	51.154	(305.734)	37.7	2,473	4,360
Per capita household consumption, 12m	2996.00	-34.969	(484.323)	29.2	552	1,217
Total household earnings, 12m	1473.14	432.564	(339.135)	36.0	1,171	2,133
Among males:						
Total non-farm earnings, 12m	3123.12	-176.775	(536.779)	30.2	2,343	5,397
Farming profits, 12m	33.98	-36.752*	(23.677)	31.2	1,245	2,760
Self-employment profits, 12m	719.90	-32.355	(178.469)	33.8	2,604	5,445
Wage earnings, 12m	2190.84	122.316	(468.680)	31.3	2,412	5,385
Per capita household consumption, 12m	4183.60	-103.192	(846.211)	29.1	635	1,470
Total household earnings, 12m	2320.42	156.744	(597.133)	28.2	1,044	2,539

Notes: Monetary outcomes are expressed in 2024 USD PPP and trimmed of the top one percent within each survey round. All outcomes cover the last 12 months. Estimates pool observations across KLPS rounds, with availability varying by outcome (total non-farm, self-employment, and wage earnings in KLPS-2 through KLPS-5; farming profits and per-capita consumption in KLPS-3 and KLPS-4; household earnings in KLPS-4 and KLPS-5), include survey-round indicators as covariates (and, in the pooled panel, an indicator for female respondents), and cluster standard errors by respondent. *Total non-farm earnings* is the sum of wage earnings across all jobs and non-agricultural self-employment profits across all businesses. *Farming profits* is net revenue from farming for which the respondent supplied the household labor and was the main decision-maker, less input costs. *Per capita household consumption* is the value of food and non-food consumption divided by household size, with an urban price adjustment for Nairobi and Mombasa. *Total household earnings* is the sum of wage earnings, self-employment profits, and agricultural profits across household members, divided by household size. KCPE scores are re-centered at the gender-specific cutoffs (234 for females, 251 for males). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the outcome mean among those scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A12: Characteristics of Assessed Children

		Reduced form				
	Below-cutoff mean	Estimate	(SE)	Band-width	Eff. obs.	Obs.
Children of female respondents:						
Child is female	0.51	-0.093	(0.118)	34.1	465	943
Child's age at assessment (years)	11.05	-0.297	(0.442)	37.4	518	943
Older child selected	0.54	-0.092	(0.125)	29.7	404	943
Parent's age at child's birth	28.29	1.078*	(0.648)	31.7	394	866
Number of children selected for assessment	1.47	0.004	(0.102)	42.1	581	943
Children of male respondents:						
Child is female	0.50	0.127	(0.112)	33.2	539	1,102
Child's age at assessment (years)	11.08	-0.364	(0.456)	32.7	524	1,102
Older child selected	0.53	-0.077	(0.112)	34.0	554	1,102
Parent's age at child's birth	29.10	0.441	(0.682)	35.3	537	1,025
Number of children selected for assessment	1.56	-0.092	(0.101)	36.4	581	1,102

Notes: Observations are children aged 8–14 assessed in the KLPS-5 child module. Panels split the sample by the *parent's* sex, with KCPE scores re-centered at the parent's gender-specific cutoff (234 for females, 251 for males). *Older child selected* indicates the older of the two children selected for assessment. *Parent's age at child's birth* is the child's birth year minus the parent's year of birth. *Number of children selected for assessment* is the number of the respondent's children selected (one or two). Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel; *Bandwidth* is the MSE-optimal bandwidth in KCPE points, *Eff. obs.* the observations within it, and *Below-cutoff mean* the mean among children of parents scoring below the cutoff. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

Table A13: Children's Academic Skills with Controls

	(1)	(2)	(3)
Children of female respondents:			
Academic index	0.362*	0.410**	0.371*
	(0.212)	(0.207)	(0.210)
Language index	0.414*	0.459**	0.395*
	(0.243)	(0.243)	(0.246)
Observations	943	943	866
Children of male respondents:			
Academic index	-0.172	-0.180	-0.180
	(0.247)	(0.234)	(0.240)
Language index	-0.086	-0.090	-0.056
	(0.225)	(0.211)	(0.208)
Observations	1,102	1,102	1,025
Child controls		X	X
Parent's age at child's birth			X

Notes: Observations are children aged 8–14 assessed in the KLPS-5 child module. Panels split the sample by the *parent's* sex, with KCPE scores re-centered at the parent's gender-specific cutoff (234 for females, 251 for males). Indices are standardized to mean zero and unit variance in the full KLPS-5 child sample. *Child controls* are the child's sex, age at assessment in months, an indicator for the older of the two selected children, and an indicator for survey wave 2. Column (3) also controls for the parent's age at the child's birth and excludes children with a missing birth year. Estimates are conventional local-linear RD coefficients from `rdrobust` with a triangular kernel and MSE-optimal bandwidths. Standard errors (in parentheses) and significance stars are based on the robust bias-corrected inference of Calonico et al. (2014). * denotes significance at the 10 percent level, ** at the 5 percent level, and *** at the 1 percent level.

B Matching Approach

This appendix describes the procedure used to link individual records from the Kenya Life Panel Survey (KLPS) to the Kenya Certificate of Primary Education (KCPE) administrative data. The KLPS dataset contains detailed survey information on pupils, including multiple reported names across survey rounds and records of the primary schools they attended. The KCPE data contain the official examination records with candidate names, schools, years, and exam scores. The procedure is strict, with the goal of linking each KCPE record to a single KLPS respondent based on consistency between reported names and schools.

Step 1: Assembling identifying information. The first step compiles all available name and school variables from the KLPS surveys. In each KLPS round, respondents are asked to report their name again, and across rounds they often provide slightly different versions or spellings. All names ever reported in the KLPS are used, to maximize the chances of identifying a correct match in the KCPE records. The KLPS also collects information on schooling history, including the primary schools attended, which is used alongside names. On the KCPE side, the procedure uses the candidate's full name, the school at which they sat for the exam, and the year of examination. Using the full history of reported names and schools captures as much identifying information as possible, since spelling and reporting can vary over time.

Step 2: Cleaning and standardizing names. All names are first converted to uppercase and stripped of punctuation and extraneous whitespace. Beyond these generic edits, specific replacement rules standardize common variants, such as "Maureen/Morine/Maurine" to MAUREEN, "Bernard/Benard" to BERNARD, and "Jackline/Jacqueline" to JACKLINE. Common regional variants of surnames are also harmonized, for example treating ODUOR, ODUAR, and ODOUR as the same name, standardized to ODUOR. Each name is then broken into tokens (for example, "John Paul Otieno" becomes {John, Paul, Otieno}), which allows matches across different name orders and combinations.

Step 3: Cleaning and standardizing schools. A similar cleaning process is applied to school names. Suffixes such as "Primary School," "PS," or "PRI SCH" are removed, and frequent transcription mistakes are corrected. When the same school appears with multiple spellings or abbreviations, these are collapsed into a single standardized form.

Step 4: Restricting matches by school. The first filter requires that the survey data

and KCPE records agree on school. Using the cleaned and standardized school names, each survey school is matched exactly to the KCPE schools, blocking on exam year: candidate pairs are formed either when the school the child reported attending in a given year matches the KCPE school for that year, or, more broadly, when any school ever reported for the child matches the KCPE school and the exam year falls within the child's plausible KCPE year range. This step limits the pool of potential matches for each child to candidates who sat for the KCPE at the same school.

Step 5: Matching names within schools. Within the restricted set of schools, the cleaned and tokenized names from the KLPS and KCPE are compared. Overlap is measured between the sets of tokens, allowing for small differences in spelling using fuzzy string matching: tokens are considered a match at a Jaro–Winkler similarity of 0.86 or higher. Candidate matches are then scored on two factors: (1) the share of KCPE tokens that appear in the survey names, and (2) whether there is an exact or close school match. The scoring is designed so that full coverage of the KCPE name tokens combined with school agreement yields the maximum possible score.

Step 6: Identifying perfect matches. A match is considered "perfect" if the survey names capture all tokens in the KCPE record and the school matches. These are high-confidence matches. If a respondent has exactly one perfect match, it is accepted automatically. When a respondent has multiple perfect matches, additional filters are needed to resolve them.

Step 7: Applying additional filters for multiple matches. Two further checks are applied. First, matches in which the KCPE record lists a sex that directly contradicts the survey information are dropped. Second, the exam year is checked for consistency with the respondent's reported grade progression in the KLPS. Specifically, the expected KCPE year is estimated from the grade the child reported being in at baseline and compared with the official exam year. Matches that are implausibly early are removed, and if more than one plausible record remains, the earliest KCPE year consistent with the child's schooling trajectory is selected.

Step 8: Resolving collisions across respondents. In some cases, multiple KLPS respondents match the same KCPE record. These "collisions" require additional adjudication, which compares how closely each respondent's expected exam year aligns with the actual KCPE year. If only one respondent falls within a small acceptable margin (within zero to

two years), that respondent is selected.

C Additional Details on the Validity of the Design

Stability of the cutoffs over time. Because both the KCPE scale and the examination itself changed across years, Appendix Figure A3 repeats the structural break search separately for exam years 1999–2000, 2001–2002, and 2003–2006. These searches are much noisier than the pooled search, because each gender-by-period cell contains only about 280 to 620 observations within the search window, compared with about 1,100 to 1,400 in the pooled search, and the resulting R^2 profiles are correspondingly flat. Within the range of 225 to 275 points, the female search peaks at 234 in 1999–2000 and 2001–2002 and at 239 in 2003–2006, and the male search peaks at 251 in 1999–2000, and at 243 and 227 in the later periods. In three of the six searches (women in 1999–2000 and men in 2001–2002 and 2003–2006), the R^2 is highest outside this range, at 213, 223, and 298 points. In each of these cases the profile is nearly flat, and the maximum R^2 exceeds the highest value within the range by at most 0.012, so these subsamples are uninformative about the location of the threshold rather than pointing to a different one.

Match rates. Appendix Table A4 estimates the discontinuity in the probability that a KCPE archive record is matched to a KLPS respondent. The level of this match rate is mechanically low, because the archive includes candidates from other cohorts and repeat attempts that could not be linked to the KLPS sample. Even if every KLPS respondent who reached Standard 8 were linked, at most 30% of records could be matched, and just below the cutoff the procedure attains about three-quarters of that ceiling. The object of interest is therefore the discontinuity rather than the level. Match rates are smooth at the cutoff in the pooled sample and among male examinees, but fall by 5 to 6 percentage points among female examinees. Retaking was common in this cohort. Among the respondents Ozier (2018) matched to official records, about 10% had scores from two exam years, typically consecutive. Near-miss candidates who retook the exam could cross the threshold on their second attempt. Because the matching procedure links each respondent to the earliest exam record consistent with her schooling history, these later attempts appear in the archive as unmatched records concentrated just above the cutoff. Consistent with this interpretation, apparent repeat records are almost never matched (2.5%, compared with 19% overall) and lie overwhelmingly above the cutoff, and the female discontinuity shrinks and loses significance when the score points nearest the threshold are excluded (donut estimates of -0.024 to -0.039, with robust p-values of 0.18 to 0.40). Ozier (2018) found a related deficit in the density of matched women just above the cutoff (McCrary $p = 0.017$) and noted it as a reason for caution in interpreting the results for women. In the expanded sample used here, the density of matched respondents is smooth at the cutoff for both women ($p = 0.92$) and

men ($p = 0.74$), and covariates are balanced among matched women (Appendix Table A2).